How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

by

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A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Science

in

Statistics

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Abstract

Task descriptions and demonstrations both convey information about a task, but their relative value is difficult to quantify. We define the predictive effective sample size (ESS) of a description as the number of additional demonstrations required to match its reduction in prediction error. In a Bayesian model of in-context linear regression, we distinguish a description's specificity—the precision of the task information it provides when relevant—from its reliability—the probability that this information applies. Reliable descriptions can replace many demonstrations, with a benefit that depends on task dimension as well as precision. Any fixed positive probability of irrelevance, however, bounds their ESS when no demonstrations are available. Transformers meta-trained on this task approach the Bayesian benchmark and learn how much to trust descriptions from their training distribution. We then test pretrained language models on numerical prediction tasks with explicit probabilistic descriptions and on a semantic classification task. Without reasoning, the models respond weakly to stated precision; with reasoning, several approach the Bayesian reference. With reasoning, they also update a description's reliability from the demonstrations when the prompt states what follows if the description is wrong, but not when it leaves this open. Both behaviors depend on how explicitly the prompt specifies the description's statistical meaning. ESS thus provides a common scale for the information a description offers and the benefit a learner obtains from it.

Preface

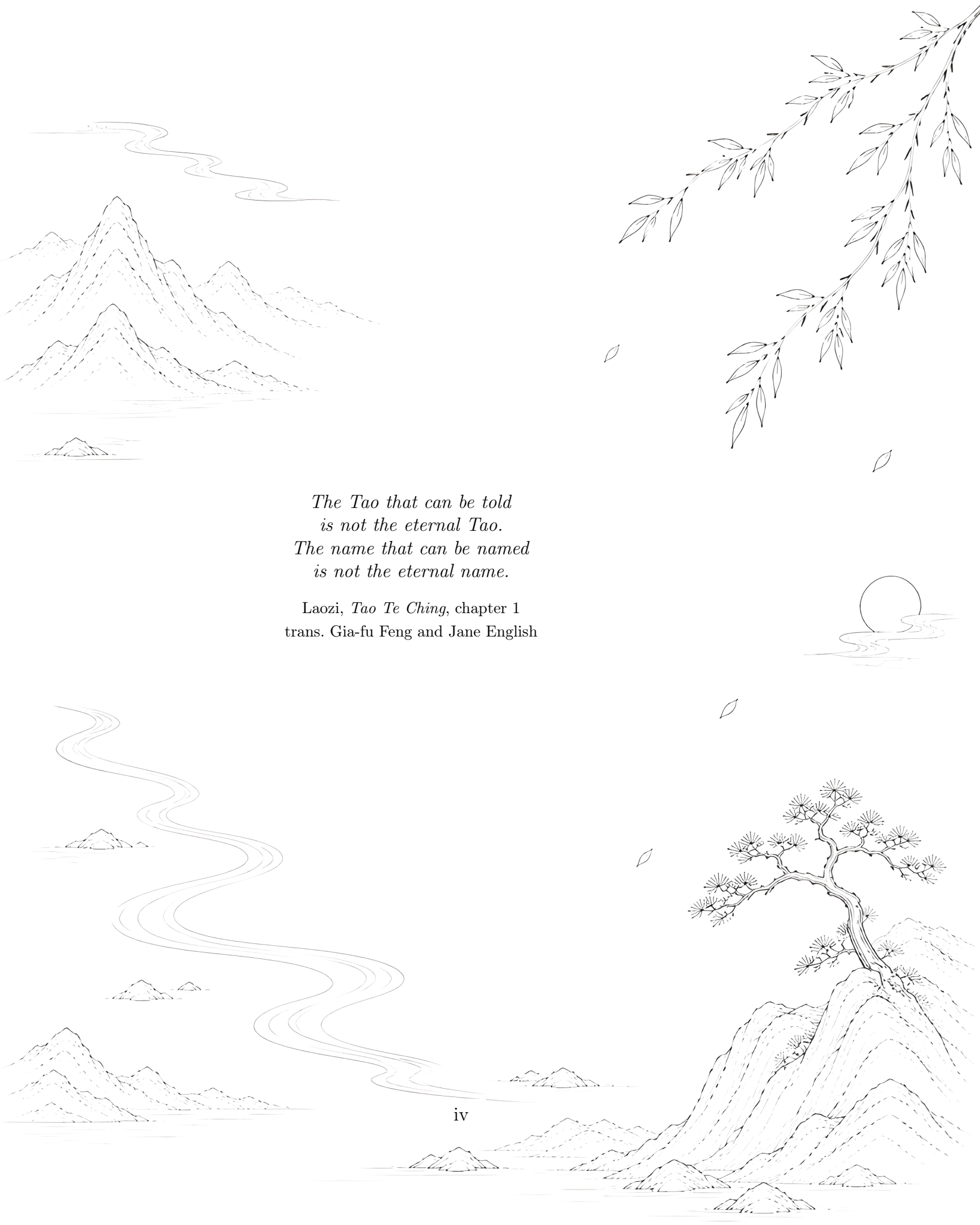
This thesis is an original work by Trung Quan Nguyen, carried out under the supervision of Professor Linglong Kong and Professor Bei Jiang. **[TODO: Describe any contribution of the supervisors or others to the research questions, design, analysis, or writing, or state that there was none.]**

The thesis is based on the manuscript "How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning" by Bruce Quan Nguyen, the name under which the author writes, prepared in the format of the International Conference on Machine Learning. Chapters 1 and 3 to 8 and Appendix A follow that manuscript's text, with its appendices placed in the chapters they support and some added detail; Section 8.2 is new. Chapter 2 is new, except that Sections 2.2, 2.5, 2.8, and 2.9 take the manuscript's review of related work. **[TODO: State whether the manuscript has been submitted or published; if so, give the full citation.]**

The research involved no human participants, animals, or personal data, so no research ethics approval was required.

Large language models are the object of study in Chapter 7. Separately, generative AI assistants were used in carrying out this research. **[TODO: Confirm and adjust this draft: Anthropic's Claude Code and OpenAI's Codex were used to write and run experiment code, to check derivations, and to draft and edit text; the author designed the study and reviewed, and takes responsibility for, all code, results, and text. Confirm that the program and supervisory committee permitted this use.]**

[TODO: Acknowledge third-party funding, such as a supervisor's grant, a scholarship, or Decaude Labs, and any paid compute; or state that the work was unfunded.]



The Tao that can be told
is not the eternal Tao.
The name that can be named
is not the eternal name.

Laozi, *Tao Te Ching*, chapter 1
trans. Gia-fu Feng and Jane English

Acknowledgements

I thank my supervisors, Professor Linglong Kong and Professor Bei Jiang, for their guidance and support throughout my program. **[TODO: Thank the examining committee, colleagues, funders, and family as you wish.]**

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Chapter 1

Introduction

An in-context learner can learn about a task in two ways: a prompt can describe what to do, or it can demonstrate the desired behavior. Most practical prompts combine the two (Brown et al., 2020), and task descriptions also appear in system instructions and the files used to configure coding agents (Chatlatanagulchai et al., 2025). Yet we lack a common scale for comparing the information supplied by a description with that supplied by examples. When should a precise description replace many demonstrations? When can examples help a learner recognize that a description does not apply?

Empirical findings suggest that the answers depend on both the description and the learner. Prompts can substitute for substantial labeled data in prompt-based fine-tuning (Le Scao & Rush, 2021), and written summaries can replace demonstrations (Honda et al., 2025). Other studies find weak sensitivity to instruction relevance (Webson & Pavlick, 2022), excessive reliance on demonstrations relative to explicit instructions (Gupta et al., 2025), or limited benefits from repository instruction files (Gloaguen et al., 2026). These settings differ, but they raise a shared measurement problem: how much task information does a description provide, and how much of it does the model use?

We address this problem within the Bayesian view of in-context learning (ICL), which treats the prompt as evidence about a latent task (Xie et al., 2022). A description induces a distribution over tasks before the demonstrations are observed. We characterize this distribution through two properties. *Specificity* measures how tightly a relevant description constrains the task. *Reliability* measures the probability that the description is relevant in the first place. A broad but correct description and

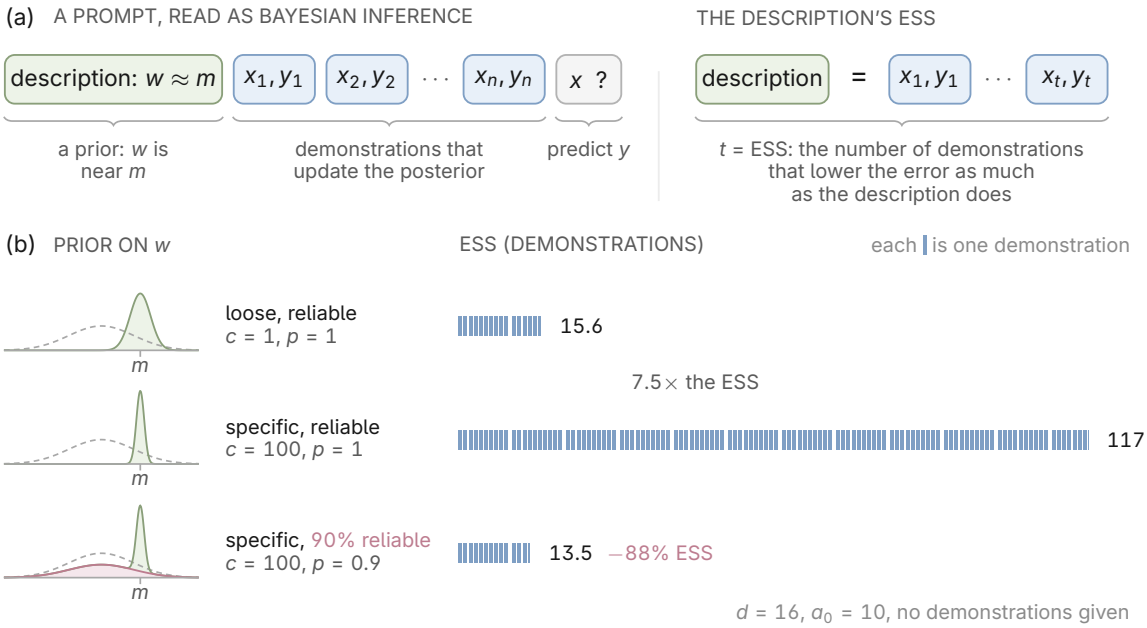


Figure 1.1: **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w; demonstrations supply additional evidence. Its effective sample size (ESS) is the number of additional demonstrations needed to achieve the same prediction error. (b) Baseline and description-conditioned task distributions, shown schematically, with their Bayesian ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Specificity c is the precision of a relevant description; reliability p is its probability of relevance. Pink denotes an irrelevant description. Increasing specificity can greatly increase ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit without demonstrations.

a precise description of the wrong task create different inference problems: the latter requires the learner to assess whether the demonstrations support the description.

To compare descriptions with demonstrations, we adapt the effective sample size of an informative prior (Morita et al., 2008; Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021). Our *predictive ESS* is the number of additional demonstrations that a learner without the description needs to match the learner's error with it. The definition applies at any number of demonstrations and to any predictor. It therefore distinguishes the benefit available to a Bayesian learner from the benefit realized by a trained model.

We develop this framework in Gaussian linear regression, where description informa-

tion and prediction error can be analyzed directly. We then study two ways a learner can acquire that information. Meta-trained Transformers learn the specificity and reliability of description tokens from their training distribution. Pretrained language models instead receive explicit probabilistic descriptions in text. This second setting isolates whether models use stated uncertainty when combining a description with numerical demonstrations; it does not require assigning an exact probability distribution to an unrestricted natural-language instruction.

1.1 Contributions

Our contributions are threefold:

- **A predictive measure of description value.** We define description ESS by matching prediction errors, using each learner's own learning curve as the reference (Definition 3.1).
- **An analysis of specificity and reliability.** We derive risk formulas and ESS limits for Gaussian task descriptions. Reliable descriptions benefit from both precision and coverage of task dimensions. Unreliability imposes a finite ceiling without demonstrations, while demonstrations can help assess relevance (Chapters 4–5).
- **Tests of learned and stated task information.** Meta-trained Transformers approach the Bayesian benchmark. In pretrained language models, reasoning substantially improves the use of stated precision, and recovers Bayesian reliability updating when the prompt states what a wrong description implies, in both numerical and semantic tasks. Both depend on how explicitly the prompt specifies the description's statistical meaning (Chapters 6–7).

1.2 Outline

The results of Chapters 3 to 7 also appear in the manuscript described in the Preface. This thesis presents them in full, together with the background they rest on, complete proofs, and all experiments. The rest of the thesis is organized as follows. Chapter 2 reviews the background, namely in-context learning and its Bayesian interpretation, the effective sample size of a prior, and robust mixture priors, together with related work. Chapter 3 presents the framework: tasks and demonstrations, descriptions as

uncertain task information, prediction error, and predictive ESS. Chapter 4 derives the value of a reliable description, with its proofs and the analysis of partial descriptions. Chapter 5 derives the cost of unreliability, with its proofs and the analysis of the fully trusting predictor. Chapter 6 tests meta-trained Transformers, with their training details and further networks. Chapter 7 tests pretrained language models on numerical prediction tasks with explicit probabilistic descriptions and on a semantic classification task, with the evaluation details and additional results. Chapter 8 discusses the findings and their limitations, and outlines future work. Appendix A reports the language-model tables at every evaluated demonstration count.

Chapter 2

Background

In this chapter, we review the ideas that the rest of the thesis builds on. Sections 2.1 to 2.3 introduce in-context learning, its interpretation as Bayesian inference, and in-context linear regression, a setting in which that interpretation can be computed exactly. Section 2.4 defines the squared error by which we compare learners and gives the error of the Bayesian predictor. Sections 2.5 and 2.6 present two tools from Bayesian statistics that we later apply to descriptions: the effective sample size of a prior and the robust mixture prior. Finally, Sections 2.7 to 2.9 review the work most closely related to ours.

2.1 In-Context Learning

A language model takes a sequence of tokens, the prompt, and predicts what comes next. Brown et al. (2020) showed that large language models can perform a task specified in the prompt, without any update to their weights. This ability is called in-context learning (ICL). Brown et al. (2020) distinguished three kinds of prompts. A zero-shot prompt gives only a natural-language description of the task. One-shot and few-shot prompts add one or several *demonstrations*, solved examples of the task, before the query that the model must answer. We follow their terminology. A demonstration is one input together with its label, and apart from the query we call everything else in the prompt a *description*, whether it is written as a statement about the task or as a command. A prompt can contain both, and our question is how much the description adds to the demonstrations.

How models use the demonstrations is itself an active question. Pretrained models can perform well even when the labels of the demonstrations are replaced by random ones (Min et al., 2022), which suggests that they use demonstrations partly to recognize a task they already know. Larger models, however, follow flipped in-context labels more readily (Wei et al., 2023). Lin & Lee (2024) explain both behaviors, retrieving a known task and learning a new one, within a single Bayesian model. A description raises the same question from the other side. Does a model treat it as information about the task, to be weighed against the demonstrations, or does it use it in some other way?

2.2 In-Context Learning as Bayesian Inference

Xie et al. (2022) proposed that ICL can be understood as implicit Bayesian inference. Suppose that pretraining text is generated from latent concepts. A model that predicts such text well must then infer the concept behind each document, and a prompt of demonstrations is evidence about the concept behind the prompt. In this view, a task is a latent parameter w, and a prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x. The ideal learner predicts the query's label with the posterior predictive distribution

$$f(y \mid x, D_n) = \int f(y \mid x, w)\, f(w \mid D_n)\, dw. \tag{2.1}$$

This view becomes testable when the prior over tasks is known. Consider a Transformer trained on prompts whose tasks are sampled from a known prior. Training on such prompts is called meta-training: the network is trained on many tasks and must infer from the context which task a prompt comes from. The posterior predictive minimizes the expected log loss over the prior (Section 2.4), and this expected log loss is exactly the training objective. A well-trained network therefore approximates the posterior predictive, whose mean is the Bayes-optimal prediction under squared error (Section 2.4) (Müller et al., 2022). For regression with a Gaussian task prior, meta-trained Transformers behave like the Bayesian predictor, which is ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024), provided that they are trained on enough distinct tasks (Raventós et al., 2023). The Bayesian predictor is then a benchmark against which a network can be measured, and Chapter 6 uses it in this way.

Bayesian accounts of ICL. ICL can be viewed as inference about a latent task (Xie et al., 2022), and Transformers trained on task distributions can approximate posterior prediction (Müller et al., 2022; Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). Work incorporating additional task information considers input-distribution descriptions (Huang & Ge, 2025), candidate hypotheses (Lin et al., 2025), instructions in generalization bounds (Tong et al., 2026), and prefixes of related datasets for hierarchical Bayesian prediction (Zhu et al., 2026). We focus on a common quantitative question across such settings: how much does task information reduce the number of demonstrations needed for a given prediction error?

2.3 In-Context Linear Regression

Garg et al. (2022) introduced in-context linear regression as a controlled test of ICL. Each prompt has its own task vector $w \sim \mathcal{N}(0, I_d)$, its inputs are drawn as $x_i \sim \mathcal{N}(0, I_d)$, and its labels are $y_i = w^\top x_i$. Garg et al. (2022) found that a Transformer trained from scratch on such prompts predicts the query's label about as well as least squares. Later work added label noise, $y_i = w^\top x_i + \epsilon_i$ (Akyürek et al., 2023; Raventós et al., 2023). Raventós et al. (2023) also varied the number of distinct tasks seen in training. With few tasks, the network behaves like the Bayesian predictor for a prior over those tasks alone. With enough tasks, it behaves like ridge regression, the Bayesian predictor for the Gaussian prior.

With Gaussian noise and a Gaussian prior, Bayes-optimal prediction has a closed form (Gelman et al., 2013, Section 14.8). We measure labels in units of their noise, so that $y_i \sim \mathcal{N}(w^\top x_i, 1)$. Let the prior be $w \sim \mathcal{N}(m, \Lambda^{-1})$, with center m and precision Λ. Let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and write $W_n = X_n^\top X_n$. The posterior over w is Gaussian, with covariance and mean

$$\Sigma = \big(\Lambda + W_n\big)^{-1}, \qquad \mu = \Sigma\big(\Lambda m + X_n^\top y_{1:n}\big), \tag{2.2}$$

and the posterior predictive for the query is

$$y \mid x, D_n \sim \mathcal{N}\big(\mu^\top x,\ 1 + x^\top \Sigma x\big). \tag{2.3}$$

The predictive variance has two parts. The first, 1, is the label noise. The second, $x^\top \Sigma x$, is the remaining uncertainty about the task in the direction of the query. Under

the baseline prior $\mathcal{N}(0, a_0 I_d)$, the posterior mean $\mu = (W_n + a_0^{-1} I_d)^{-1} X_n^\top y_{1:n}$ is the ridge regression estimate with penalty a_0^{-1}. Equation (2.2) also shows that a prior acts like additional data. Each demonstration adds $x_i x_i^\top$ to the posterior precision, and the prior adds Λ. Chapter 3 uses this fact to give a description a prior of its own, and in Chapter 4 a reliable description replaces the baseline precision a_0^{-1} with c.

2.4 Squared Error

A predictor sees the prompt and gives a number $\hat{y}$ for the query's label. We score it by squared error. The label is $y = w^\top x + \epsilon$, where the noise ϵ has mean 0 and variance 1 and is independent of the prompt and the task. The expected squared error therefore splits into two parts:

$$\mathbb{E}\big[(\hat{y} - y)^2\big] = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big] + 1. \tag{2.4}$$

The second part is the noise variance, which is the same for every predictor. Following Garg et al. (2022), we therefore score a predictor by the first part alone, its *error* or *risk*

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big],$$

the expected squared distance from the noiseless label $w^\top x$. A predictor that knows the task predicts $w^\top x$ and has zero error.

Given the prompt, the task is uncertain. For any prediction $\hat{y}$ computed from the prompt,

$$\mathbb{E}\big[(\hat{y} - w^\top x)^2 \mid x, D_n\big] = \big(\hat{y} - \mathbb{E}[w^\top x \mid x, D_n]\big)^2 + \mathrm{Var}(w^\top x \mid x, D_n).$$

The first term vanishes only when $\hat{y}$ is the posterior mean of $w^\top x$. When tasks are drawn from the prior, the posterior mean therefore has the smallest error of any predictor. We call it the Bayesian predictor and its error Bayes-optimal. It is also the mean of the posterior predictive (2.1). Its error is the expected posterior variance of $w^\top x$. In linear regression, the posterior is Gaussian with covariance Σ (Equation (2.2)), so the posterior variance of $w^\top x$ is $x^\top \Sigma x$, the part of the predictive variance (2.3) that comes from uncertainty about the task. The query input is independent of the demonstrations, with $\mathbb{E}[xx^\top] = I_d$, so the error of the Bayesian predictor is

$$R = \mathbb{E}\big[x^\top \Sigma x\big] = \mathbb{E}\,\mathrm{tr}\,\Sigma, \tag{2.5}$$

the expected trace of the posterior covariance. The error is zero when the posterior has concentrated on the task, and it grows with the uncertainty about the task that remains.

Two standard facts give this trace in the cases that later chapters need. Chapter 3 writes $R^{\mathrm{demo}}(n)$ for the error with n demonstrations and no description. With fewer demonstrations than dimensions, $n < d$, the demonstrations say nothing about $d - n$ directions, so under the baseline prior $\mathcal{N}(0, a_0 I_d)$ we have $R^{\mathrm{demo}}(n) \geq (d-n)\, a_0$. Under a flat prior, the limit $a_0 \to \infty$, and with $n \geq d + 2$, the posterior covariance is W_n^{-1}, the inverse of a Wishart matrix, and the error is $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = d/(n - d - 1)$ (Muirhead, 1982). Chapter 4 builds its limits for a reliable description (Proposition 4.1) from these two facts.

A predictor that gives a density $g(y \mid x, D_n)$ rather than a number can instead be scored by its log loss, $-\log g(y \mid x, D_n)$. The expected log loss is smallest for the true conditional density of y given the prompt (Cover & Thomas, 2006), which is the posterior predictive when tasks are drawn from the prior. Meta-trained networks are often trained with this loss (Section 2.2), and so are the networks of Chapter 6. The mean of the posterior predictive is the posterior mean, so a network that minimizes log loss and outputs a predictive mean can be scored by squared error. We use squared error throughout for two reasons. First, it needs only the learner's answer, a number, so it applies to a language model through the number that it writes (Chapter 7). Second, the Bayes-optimal error (2.5) has the exact forms above, so the theory in Chapters 4 and 5 can work with exact errors.

Squared error also splits exactly over the components of a mixture. Suppose that a latent indicator Z selects the component that generates the task. By the law of total variance, the posterior variance is the sum of two parts:

$$\begin{aligned} \mathrm{Var}(w^\top x \mid x, D_n) = \mathbb{E}\big[\, \mathrm{Var}(w^\top x \mid x, D_n, Z) \mid x, D_n \big] \\ + \mathrm{Var}\left(\mathbb{E}[w^\top x \mid x, D_n, Z] \mid x, D_n\right). \end{aligned} \tag{2.6}$$

The first part is the average variance within the components. The second is the variance of the prediction across the components. Chapter 5 applies this decomposition to a description that may be irrelevant. There, the first part gives the error of an oracle that knows Z, and the second is the cost of uncertainty about relevance.

2.5 The Effective Sample Size of a Prior

Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information. In the conjugate normal model, this number is exact. Suppose that the observations are $y_i \sim \mathcal{N}(w, \sigma^2)$ and the prior is $w \sim \mathcal{N}(m, \sigma^2/c)$. After n observations with mean $\bar{y}$, the posterior is

$$w \mid y_{1:n} \sim \mathcal{N}\Big(\frac{c\,m + n\bar{y}}{c+n},\ \frac{\sigma^2}{c+n}\Big).$$

This is the posterior that $c+n$ observations would give under a flat prior if c of them averaged m. The prior is therefore worth c observations, which is its precision measured in units of the precision of one observation. Chapter 7 uses this model for its prompt-based Bayesian reference, ignoring negligible boundary effects. The specificity c of a description, defined in Chapter 3, is the same conjugate-prior ESS, counted in each direction in units of direct observations of the task. Chapters 4 and 5 show how the ESS of a description, counted in demonstrations, differs from it.

Prior effective sample size. ESS is an established way to quantify informative priors (Morita et al., 2008; Neuenschwander et al., 2020). Our measure builds on risk-based definitions (Reimherr et al., 2021) and depends on the observations already available, as does effective current sample size (Wiesenfarth & Calderazzo, 2020). Related work discounts incompatible historical information using predictive loss (Reznik, 2026); robust priors similarly account for possible mismatch (Pichon et al., 2012; Schmidli et al., 2014). Selby et al. (2025) evaluate priors elicited from language models by their predictive sample value. We instead measure descriptions supplied to an in-context learner and compare the value predicted by a task model with the value realized by the learner.

These definitions extend the ESS beyond the conjugate case in different ways. Morita et al. (2008) extend it to any parametric prior. Their ESS is the number of hypothetical observations that, starting from a very vague prior, would give a posterior carrying about the same information as the prior, where information is measured by the curvature of the log density. Neuenschwander et al. (2020) require an ESS to be predictively consistent: on average, the ESS of the posterior after N observations should equal the prior's ESS plus N. Other definitions let the ESS depend on the current sample, through its size or through the observed data themselves. Reimherr

et al. (2021) compare a prior with a baseline prior. Their ESS is the number of observations that the baseline would need to perform as well at a given sample size, and it can be negative when the prior and the likelihood conflict. Similarly, the effective current sample size of Wiesenfarth & Calderazzo (2020) counts the observations that must be added to, or removed from, the current sample for an analysis without the prior to match the analysis with it. Finally, Reznik (2026) measures information by predictive log loss to choose the discount of a power prior. A power prior raises the likelihood of historical data, which may not match the current data, to a power between 0 and 1.

Our predictive ESS (Definition 3.1) is the number of additional demonstrations that a learner without the description needs to match the learner's error with it. Like the measure of Reznik (2026), it measures information by predictive loss, here squared error.

Our ESS also differs from that of Morita et al. (2008) in what it matches. Their ESS matches information. In the conjugate normal model above, a prior of precision c adds c to the posterior precision whatever the number of observations, so it is worth c observations at every sample size. In the same way, a description that raises the prior precision from a_0^{-1} to c in every direction is worth $c - a_0^{-1} = (1 - r)\, c$ by their measure, whatever the number of demonstrations in hand. Our ESS matches errors instead. In d dimensions, n demonstrations leave the Bayesian predictor an error of $d/(n - d - 1)$ under a flat prior (Section 2.4), not the d/n that their expected information would give. Matching errors therefore adds a term that grows with the dimension, the dimension term of Chapter 4. This term fades as demonstrations accumulate, and with many demonstrations in hand the ESS_n of a reliable description returns to their value (Corollary 5.2(ii)).

2.6 Robust Mixture Priors

A prior built from historical data can conflict with the current data. Evans & Moshonov (2006) check for such prior–data conflict by asking whether the observed data are surprising under the prior predictive distribution. Clinical trials that borrow historical control data protect against conflict with a robust mixture prior (Schmidli et al., 2014). The informative prior f_I, derived from the historical data, is mixed with a vague prior

f_V:

$$f(w) = p\, f_I(w) + (1 - p)\, f_V(w).$$

The posterior is then a mixture of the two component posteriors, and the weight on the informative component is updated in proportion to p times its marginal likelihood. When the data agree with the informative component, its weight grows, and the trial borrows the historical information. When they conflict, the weight moves to the vague component, and the analysis falls back on the current data.

A description is a prior of the same kind. It may describe the task at hand, but it may also have been written for another task, or it may simply be wrong. Chapter 3 therefore models a description as a robust mixture prior whose weight is its reliability p, the probability that the description is relevant. Chapter 5 writes the updated weight as π_n and shows how reliability affects the description's ESS.

2.7 Descriptions in Theories of ICL

A few theories of ICL include descriptions (Section 2.2). In some, the description is a set of tokens that indicate a hypothesis class (Lin et al., 2025) or carry the mean of the inputs (Huang & Ge, 2025). In others, instructions enter a generalization bound (Tong et al., 2026). These theories generally do not measure the value of a description in demonstrations, and they do not model the possibility that it is irrelevant. For example, Huang & Ge (2025) show that a learner can use the description token to predict better, but they do not say how many demonstrations it is worth or what it is worth when it may be wrong. The Bayesian theory of Xie et al. (2022) models the prompt as demonstrations alone. We keep its view of the prompt as evidence about a latent task and add a description, which induces a distribution over tasks before the demonstrations are observed. Our networks in Chapter 6 use the prefix embedding of Huang & Ge (2025), in which the description is a single token placed before the demonstrations. The work closest to our setting is that of Zhu et al. (2026), who show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction. We measure the value of such conditioning information in demonstrations, and we allow it to be irrelevant.

2.8 Descriptions and Priors in Pretrained Language Models

Descriptions and evidence in language models. Controlled studies find that demonstrations can outweigh stated priors (Gupta et al., 2025). Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), and Bayesian reference curves for in-context prediction (Akata et al., 2026; Arora et al., 2025). We vary stated precision and reliability separately, use fixed-value controls, and compare reasoning modes. Further related work is discussed in Section 2.9.

In these reference comparisons, the learning curves of pretrained models approach a Bayesian reference on synthetic tasks with a known optimum (Akata et al., 2026), and Arora et al. (2025) model in-context learning curves with a Bayesian scaling law that includes a task prior. Where Min et al. (2022) and Wei et al. (2023) vary the labels of the demonstrations, we ask the converse question: does a model weigh a description according to what the prompt says about it?

2.9 Further Related Work

Descriptions as substitutes for examples. Beyond in-context learning, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect. These results motivate measuring descriptions relative to examples, while differing from our predictive-risk comparison within a fixed task distribution and learner.

Wenliang et al. (2026) treat a prompt as conditioning for a Bayesian sequence predictor and find that an optimized prompt can be shorter than demonstrations with the same effect.

Text and probabilistic prediction. Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024). Related work studies how closely language-model belief updates follow Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026). He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our experiments use a controlled

numerical task with explicit uncertainty so that description precision, relevance, and prediction weights can be compared with a specified probabilistic reference.

Models also estimate a density from numbers in context like a kernel density estimator with an adaptive bandwidth (Liu et al., 2025).

Chapter 3

Framework

Tasks and demonstrations. We use the standard noisy in-context linear regression setting (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$ drawn from the baseline prior $\mathcal{N}(0, a_0 I_d)$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x, where

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

The parameter $a_0 > 0$ controls variation across tasks. Numerical illustrations in Chapters 4–5 use $d = 16$ and $a_0 = 10$.

Descriptions as uncertain task information. A description supplies a center $m \in \mathbb{R}^d$. Its *specificity* is a precision $c \geq 1/a_0$, and its *reliability* is a probability $p \in [0, 1]$. Write $r = (ca_0)^{-1} \in (0, 1]$ for the fraction of baseline variance remaining when the description is relevant. With an unobserved relevance indicator $Z \sim \text{Bernoulli}(p)$, independent of m, we generate

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)a_0 I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d). \end{aligned}$$

Thus the description induces the mixture prior

$$w \mid m \sim p\,\mathcal{N}(m, c^{-1} I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d),$$

as in robust mixture priors for potentially incompatible information (Schmidli et al., 2014). Both relevance branches have the same marginal task distribution, $w \sim$

$\mathcal{N}(0, a_0 I_d)$. Consequently, the comparison with and without a description does not change the task population.

A description is *reliable* when $p = 1$ and *unreliable* when $p < 1$. Greater c means greater specificity conditional on relevance. The model separates uncertainty within the described region from uncertainty about whether that region applies. It does not assume that every natural-language description can be summarized by these two scalars.

A relevant description splits the baseline variance a_0 into the part $(1 - r)a_0$ that m resolves and the part ra_0 that it leaves. Because the observation noise has unit variance, c is measured in demonstrations: in each direction, it is the ESS that Morita et al. (2008) assign to a normal prior. We call a description *loose* when c is small relative to d and *specific* when c exceeds d. At $d = 16$ and $a_0 = 10$, a description with $c = 1$ ($r = 0.1$) is loose, and one with $c = 100$ ($r = 0.001$) is specific.

Prediction error. For a prediction $\hat{y}$ of the query response, we measure error against its conditional mean:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{3.1}$$

This excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, let $R^{\text{desc}}(n)$ and $R^{\text{demo}}(n)$ denote its errors with and without a description. We write $R^{\text{desc}}_{p=1}(n)$ for the reliable case.

Definition 3.1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ such that $R^{\text{demo}}(t) = R^{\text{desc}}(n)$. When this crossing exists, the ESS of the description is

$$\text{ESS}_n = n^* - n.$$

We interpolate R^{demo} between available demonstration counts: linearly in t for the theory and meta-trained models, and in $\log t$ for language models. For measured curves, we use the first crossing within the observed range and report out-of-range values as bounds. The language-model conventions are detailed in Section 7.5. If $R^{\text{desc}}(n) > R^{\text{demo}}(t)$ for all $t \geq 0$, we report $\text{ESS}_n \leq -n$.

ESS measures a horizontal displacement of a learning curve: $\text{ESS}_n = 10$ means that n demonstrations plus the description match the error of $n + 10$ demonstrations without it. For a decreasing baseline curve, a negative value indicates increased error

at the same demonstration count. Each model is evaluated against its own curve, so ESS measures the description's value *to that learner*; a large ESS alone does not imply low absolute error. Under the correctly specified Bayesian model, descriptions cannot increase expected risk, and ESS is nonnegative.

Because R averages over descriptions, the ESS belongs to a kind of description, determined by c and p, rather than to a particular value of m.

Chapter 4

The Value of a Reliable Description

For $p = 1$, the Bayesian predictor uses the posterior mean of $w^\top x$. Its risk is the expected trace of the posterior covariance of w. Let X_n contain the demonstration inputs as rows and write $W_n = X_n^\top X_n$. Then

$$R^{\mathrm{demo}}(n) = \mathbb{E}\,\mathrm{tr}(a_0^{-1} I_d + W_n)^{-1},$$
$$R^{\mathrm{desc}}_{p=1}(n) = \mathbb{E}\,\mathrm{tr}(cI_d + W_n)^{-1}.$$

The description replaces baseline precision a_0^{-1} with c. Its predictive value is not, however, determined by this precision difference alone.

Proposition 4.1 (Reliable descriptions)**.** *Let $p = 1$.*

(i) With $r \in (0, 1]$ fixed, $\mathrm{ESS}_0 \geq (1 - r)d$, and $\mathrm{ESS}_0 \to (1 - r)d$ as $a_0 \to \infty$.

(ii) With positive integer c fixed, $\mathrm{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$.

The two limits describe different regimes (proof in Section 4.1). In the first, the description removes a fixed fraction of uncertainty in every direction. Each of the first d demonstrations constrains one direction, so reducing variance across all directions can save approximately $(1 - r)d$ demonstrations. In the second, the description retains a fixed absolute precision while the baseline prior becomes diffuse. Without a description, the limiting risk is $d/(n - d - 1)$ for $n > d + 1$ (Muirhead, 1982); the description alone has risk d/c. These risks match at $n = c + d + 1$. The integer restriction makes the crossing coincide with a measured count under linear interpolation.

This $d + 1$ contribution reflects the cost of learning a d-dimensional task from random inputs. Matching average information would count only the added precision

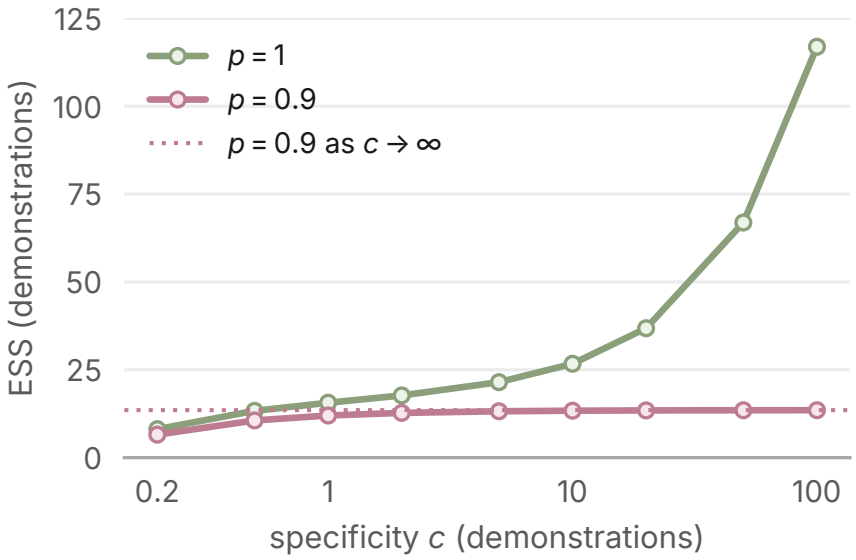


Figure 4.1: **Reliability limits the benefit of greater specificity without demonstrations.** Bayesian ESS_0 versus specificity c, for $p = 1$ and $p = 0.9$, at $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling for $p = 0.9$ as $c \to \infty$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

$c - a_0^{-1}$; matching predictive error also accounts for the random design. We refer to the additional contribution as the *dimension term*. It is substantial when c is small relative to d, whereas precision dominates when c is large. This interpretation concerns the stated limits, rather than an exact additive formula for every finite a_0 and n.

The comparison with matching information can be made explicit in this limit. For a normal prior, the ESS of Morita et al. (2008) is the ratio of the observation variance to the prior variance, so raising the baseline precision a_0^{-1} to c is worth $c - a_0^{-1} = (1 - r)c$ demonstrations under their definition, at every n. With a diffuse baseline prior, n demonstrations leave risk $d/(n - d - 1)$, not the risk d/n that their expected information would suggest. Setting d/n equal to the description's risk d/c gives $n = c$; setting $d/(n - d - 1)$ equal to it gives $n = c + d + 1$. The dimension term is the gap between the two.

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1 - r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117 (Figure 4.1). As demonstrations accumulate, the dimension benefit disappears: $\mathrm{ESS}_n \to c - a_0^{-1}$ (Corollary 5.2). For example, after 100 demonstrations the $c = 1$

Table 4.1: **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.23.

	loose, $c = 1$					specific, $c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

description is worth 1.1 demonstrations, close to its added precision of 0.9.

Coverage matters as well as precision. A description may constrain only $k < d$ directions. For any fixed precision $c > 0$, its zero-demonstration ESS approaches k as $a_0 \to \infty$; increasing that fixed precision does not remove uncertainty in the other directions. With many demonstrations, its ESS instead approaches $k(c - a_0^{-1})/d$ (Proposition 4.2). A precise partial description can therefore become more valuable as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS increases from 8.1 without demonstrations to 36 with 100 demonstrations (Table 4.1).

4.1 Reliable Descriptions

Notation and risk. Write $\varepsilon = 1/a_0$, let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These identities follow from the posterior covariance under Gaussian linear regression. Conditional on the prompt, the posterior mean minimizes squared prediction error, and the independent query satisfies $\mathbb{E}[xx^\top] = I_d$. Averaging its conditional error $x^\top \Sigma x$ therefore gives $\mathbb{E}\,\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $a_0 > 0$, $c \geq \varepsilon$, $r = 1/(ca_0) \in (0, 1]$, and $p \in [0, 1]$. Noninteger sample sizes refer to linear interpolation between the risks at adjacent integers.

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = \frac{d}{n-d-1} \quad (n \geq d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\, W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and the Bayes-optimal ESS is nonnegative.

Proof of Proposition 4.1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = ra_0 d.$$

(i) Fixed variance reduction. By (W1), $R^\varepsilon(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^{\varepsilon}(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1+a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0,1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{\varepsilon}(n) \longrightarrow \begin{cases} +\infty, & n \le d+1, \\ \dfrac{d}{n-d-1}, & n \ge d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c + d + 1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c + d + 1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c. □

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2}(\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^{\kappa}(n) = \frac{d}{n-d-1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{4.1}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. Thus a fixed change in prior precision and a fixed change in sample size affect risk at the same order. This observation underlies the large-n ESS limits below.

4.2 Partial Descriptions

A reliable description may constrain only part of a task. Suppose it leaves a fraction r of the baseline variance in $k < d$ directions and leaves the remaining $d - k$ directions unchanged. Its conditional prior precision is c in the first set of directions and $\varepsilon = 1/a_0$ in the second. Write

$$\Lambda = \operatorname{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\,\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4.2 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1 - r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.

(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - 1/a_0) = \frac{k}{d}(1 - r)c.$$

In the broad-prior limit, the unconstrained directions dominate the initial error. Even an arbitrarily precise description of $k < d$ directions therefore replaces only about k demonstrations when used alone. After many demonstrations, its value instead approaches the added precision averaged across all d directions. These two limits permit complementarity: a sufficiently precise partial description can be worth more alongside demonstrations than by itself. The numerical example in Table 4.1 shows this effect. At $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36 with 100 demonstrations.

The large-sample limit exceeds k about when $c - a_0^{-1} > d$, and this is when a partial description gains value as demonstrations accumulate. Between 0 and 100 demonstrations, a complete description at $c = 100$ falls from 117 to 109, so demonstrations substitute for a complete description and complement a partial one. At $c = 1$, ESS falls as demonstrations accumulate, whether the description is partial or complete.

Proof of Proposition 4.2. (i) No demonstrations. The description risk is

$$\operatorname{tr}\Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \geq (d-t)a_0$ gives $\mathrm{ESS}_0 \geq (1-r)k$. The normalized baseline risk converges to $(d-t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d-(1-r)k$. With c fixed, it is

$$d-k+\frac{k}{ca_0} \longrightarrow d-k.$$

Because $k<d$, both limiting levels are positive and have unique crossings, at $(1-r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda+W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda+W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\,\mathrm{tr}\,W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\,\mathrm{tr}(MB)| \leq \|M\|_{\mathrm{op}}\,\mathrm{tr}\,B$ for $B \succeq 0$, together with $(\Lambda+W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-1} \leq d\|\Lambda\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\,\mathrm{tr}(\Lambda+W_n)^{-1} = \frac{d}{n-d-1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in Corollary 5.2(ii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\mathrm{tr}\,\Lambda - d\varepsilon}{d} = \frac{k(c-\varepsilon)}{d}.$$

□

The same expansion applies to any fixed Gaussian description precision $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. Its large-sample ESS is the added precision averaged across directions, $(\mathrm{tr}\,\Lambda - d\varepsilon)/d$.

Chapter 5

The Cost of Unreliability

When $p < 1$, the learner must also infer whether the description applies. Let μ_1 and μ_0 denote the posterior means conditional on relevance and irrelevance, and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the posterior relevance probability. The Bayesian prediction is

$$\hat{y} = \big[\pi_n \mu_1 + (1 - \pi_n)\mu_0\big]^\top x.$$

Demonstrations affect both the component predictions and their relative weight.

Proposition 5.1 (Risk under uncertain relevance)**.** *For every $n \geq 0$,*

$$\begin{aligned} R^{\text{desc}}(n) = p R^{\text{desc}}_{p=1}(n) \\ + (1 - p) R^{\text{demo}}(n) + \mathcal{V}_n, \end{aligned} \tag{5.1}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

With no demonstrations,

$$R^{\text{desc}}(0) = \big[1 - p^2(1 - r)\big] a_0 d. \tag{5.2}$$

Equation (5.1) separates the error of an oracle that knows Z from the cost of uncertainty about relevance. The oracle uses the description when it applies and otherwise uses the baseline prior. The remaining term, $\mathcal{V}_n$, is the disagreement between these two predictions, weighted by uncertainty about which is appropriate. Without demonstrations, the oracle removes a fraction $p(1 - r)$ of baseline risk; a learner that must infer relevance removes only $p^2(1 - r)$. The proof follows from the law of total variance (Section 5.1).

The cost of uncertainty about relevance is small when the two predictions agree. It is also small when the demonstrations make one relevance branch far more likely than the other, because $\pi_n(1-\pi_n)$ is then near zero. Without demonstrations, the cost is $\mathcal{V}_0 = p(1-p)(1-r)a_0 d$, which grows as a more specific description pulls the two predictions apart. With demonstrations, however, a specific description is also easier to distinguish from an irrelevant one.

Corollary 5.2 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$, define $n_p^\sharp$ by

$$R^{\text{demo}}(n_p^\sharp) = (1-p^2)R^{\text{demo}}(0).$$

Then $\text{ESS}_0 < n_p^\sharp$ at every finite c, and $\text{ESS}_0 \to n_p^\sharp$ as $c \to \infty$. As $a_0 \to \infty$ with r fixed, $\text{ESS}_0 \to p^2(1-r)d$ and $n_p^\sharp \to p^2 d$. For $p = 0$, both ESS and the ceiling are zero.

(ii) With d, a_0, and c fixed, as $n \to \infty$,

$$\begin{cases} \text{ESS}_n \to c - a_0^{-1}, & p = 1, \\ \limsup_{n\to\infty} \text{ESS}_n \le p(c - a_0^{-1}), & p < 1. \end{cases}$$

Specificity cannot remove a relevance error floor. For fixed $p < 1$, Equation (5.2) gives a positive risk floor $(1-p^2)a_0 d$, even as $c \to \infty$. Hence zero-demonstration ESS has a finite ceiling. At $p = 0.9$, $d = 16$, and $a_0 = 10$, increasing c from 1 to 100 raises ESS only from 12.0 to 13.5, compared with 15.6 to 117 when $p = 1$ (Figure 4.1). The finite-a_0 ceiling is approximately 13.5; the diffuse-baseline approximation is $p^2 d = 12.96$.

An oracle that knows Z has the lower floor $(1-p)a_0 d$, which is the risk of a reliable description that leaves variance fraction $1-p$. In this setting, its ceiling is therefore 15.6 demonstrations, the ESS of the reliable $c = 1$ description; uncertainty about relevance lowers it to approximately 13.5.

With many demonstrations, ESS is at most p times that of a reliable description. Demonstrations reduce the risk when the description is irrelevant, $R^{\text{demo}}(n)$, so the term $(1-p)R^{\text{demo}}(n)$ falls and ESS can grow. In the limit, a reliable description is worth $c - a_0^{-1}$ demonstrations: the dimension benefit has disappeared, and what remains is the precision the description adds to the baseline prior. An

unreliable description is then worth at most p times as much (Corollary 5.2). After 100 demonstrations, the $c = 100$ description has ESS 109 when $p = 1$ and 86.6 when $p = 0.9$, compared with limits of 99.9 and at most 89.9. The reliable value exceeds its limit because the dimension benefit has not yet disappeared completely at $n = 100$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description that leaves variance fraction $1 - p^2(1 - r)$. For example, $c = 100$, $p = 0.9$ and a reliable description with $c \approx 0.52$ both have ESS approximately 13.5 in this setting. Their values diverge once demonstrations are available: after 100 demonstrations, the former has ESS 86.6 and the latter 0.5. Evidence can support the high-precision component of the unreliable description; the reliable low-precision description has no such component to recover.

Treating every description as relevant can be harmful. A fully trusting predictor always uses μ_1, even when the description is irrelevant. Without demonstrations, it is worse than ignoring the description when $p < 1/2$, provided the description is informative ($r < 1$). For fixed n, its risk tends to $2(1 - p)a_0 d$ as $c \to \infty$, whereas the demonstrations-only risk tends to zero as n grows. Thus a sufficiently precise, occasionally irrelevant description can outweigh useful demonstrations. At $p = 0.9$ and $c = 100$, full trust is worse than ignoring the description from 15 demonstrations onward in the reported evaluation (Section 5.2).

5.1 Unreliable Descriptions

Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input, and write $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$. Conditional on relevance, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on irrelevance, m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

The relevance probability is obtained from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query does not enter this update because it is independent of Z. Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$: the distribution of m is the same under both hypotheses. At $p = 0$ or $p = 1$, the formula reduces to the corresponding single component.

Proof of Proposition 5.1. The posterior mean prediction is

$$\mathbb{E}[w^\top x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1-\pi_n)\mu_0\big)^\top x.$$

Applying the law of total variance over Z yields

$$\begin{aligned} \mathrm{Var}(w^\top x \mid \mathcal{P}_n) = \pi_n x^\top \Sigma_1 x + (1-\pi_n) x^\top \Sigma_0 x \\ + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2. \end{aligned}$$

For any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1-p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (5.1).

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\,m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

The covariance holds because, conditional on relevance, w_i is m_i plus independent noise, and conditional on irrelevance, w_i is independent of m_i. It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (5.2). □

Proof of Corollary 5.2. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\text{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\text{demo}}(0) > (1-p^2)R^{\text{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\text{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in Section 4.1 also give

$$\text{ESS}_0 \geq p^2(1-r)d, \qquad \text{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1 - p^2$ gives $n_p^\sharp \to p^2 d$. These are broad-prior limits, not exact identities at finite a_0.

For comparison, an oracle told Z has zero-demonstration risk

$$\big(1 - p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

Its limiting ceiling is therefore pd demonstrations as $a_0 \to \infty$. The additional reduction from p to p^2 in the non-oracle expression is the cost of uncertainty about relevance.

(ii) Many demonstrations. The idea is as follows. With many demonstrations, an additional demonstration reduces risk by about d/n^2, and a prior of precision κ reduces it by κ times as much (Equation (4.1)). A reliable description raises the precision from ε to c, so it saves about $c - \varepsilon$ demonstrations. A description that is relevant with probability p saves about p times as much, and the term $\mathcal{V}_n$ can only lower its value further.

Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

Equation (4.1) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\text{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. Let $n + \eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon)=0$, the two risks coincide and $\eta_n = 0$ exactly. If $p = 1$, the description risk equals $L(n)$, so

$$\mathrm{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

If $p < 1$, the variance decomposition gives $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since the baseline risk decreases with sample size, $\mathrm{ESS}_n \leq \eta_n$, proving

$$\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-\varepsilon) = p(1-r)c.$$

The nonnegative term $\mathcal{V}_n$ is why the result is an upper bound for unreliable descriptions. □

5.2 The Fully Trusting Predictor

Without demonstrations, a predictor that always treats the description as relevant returns $m^\top x$. The moments used in the proof of Proposition 5.1 give

$$R^{\mathrm{trust}}(0) = \mathbb{E}\|w-m\|^2 = \big(1-(2p-1)(1-r)\big)a_0 d.$$

In each coordinate, these moments give $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1-r)a_0 + (1-r)a_0$. For $r < 1$, this risk exceeds the baseline $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the fully trusting predictor uses $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w-m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2-r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c\to\infty} R^{\mathrm{trust}}(n) = 2(1-p)a_0 d.$$

For every $p < 1$, this is a positive error floor at fixed n, whereas $R^{\mathrm{demo}}(n) \to 0$ as $n \to \infty$. Hence sufficiently many demonstrations can outperform full trust in a sufficiently precise unreliable description. The order of limits matters: this argument takes $c \to \infty$ at fixed n and does not assert that demonstrations can never overcome a fixed finite prior precision.

Chapter 6

Meta-Trained Transformers

We first test whether Transformers can learn to use descriptions in the setting where the Bayesian model is correctly specified. Following Garg et al. (2022), we train from scratch on linear regression tasks with $d = 5$ and $a_0 = 10$. A prompt contains an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The token does not specify c or p; the model must infer these properties from its training distribution. Half the training prompts omit the description, providing a demonstrations-only learning curve for each network. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration. Architecture, optimization, and additional experiments are reported in Section 6.1.

Matched training and test distributions. With no demonstrations, the $c = 10$, $p = 1$ networks achieve mean ESS 14.3, compared with a Bayesian ESS of 16.1 on the same prompts (Figure 6.1a). At lower reliability, and for $c = 1$, the zero-demonstration ESS is within 0.3 of the Bayesian value. The networks also capture the increased value of a partial description as demonstrations accumulate: with $k = 3$ of $d = 5$ directions described at $c = 10$, ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayesian values of 3.22 and 5.64.

Reliability shifts. We also evaluate each network at reliabilities different from its training value. We compare it with a *train-matched* Bayesian predictor, which retains the training reliability, and a *calibrated* predictor, which uses the test reliability (Raventós et al., 2023). At $n = 0$, the Bayesian prediction is $p\, m^\top x$. We therefore

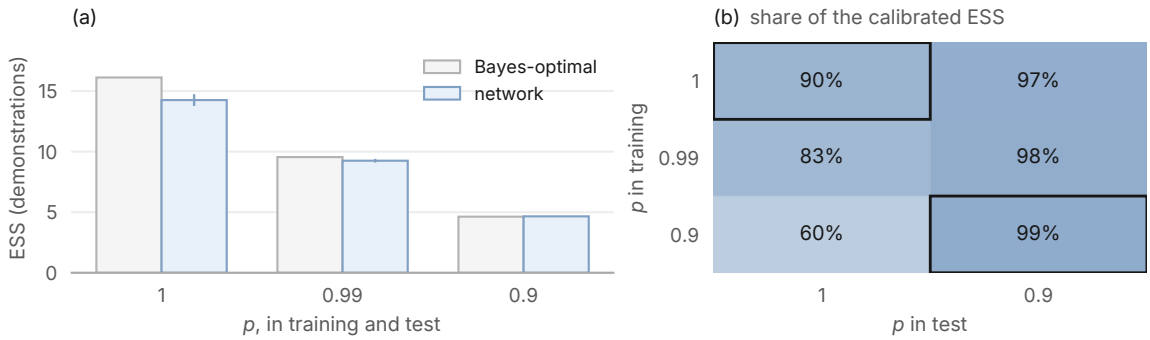


Figure 6.1: **Meta-trained Transformers approach the Bayesian ESS and inherit their trust from training.** Results for $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own demonstrations-only curve; bars show the mean and one standard deviation across two or three training seeds. The Bayesian comparison uses the same evaluation prompts. (b) Reliability changes at test time. Cells show network ESS as a fraction of the calibrated Bayesian ESS; outlined cells have matching training and test reliability. Table 6.1 also reports the $c = 1$ condition.

estimate a network's *implied trust* as the coefficient q that best approximates its predictions by $q\, m^\top x$, following coefficient-based analyses of learned predictors (Akyürek et al., 2023; Panwar et al., 2024).

For the $c = 10$ description, networks trained at $p = 0.9$ retain implied trust near 0.9 even when every test description is relevant, falling short of the calibrated predictor (Figure 6.1b). This is consistent with the information available: without demonstrations, m itself does not reveal the test reliability. Networks trained at $p = 1$ instead continue to trust descriptions fully. When 10% of test descriptions are irrelevant, their prediction error is higher with the description than without it from six demonstrations onward, in every seed. Training at $p = 0.99$ avoids this failure in the tested range: descriptions continue to reduce error at every evaluated n. Exposure to occasional irrelevance during training therefore changes how these networks combine descriptions with later evidence.

6.1 Additional Transformer Results and Training Details

Evaluation. We evaluate each network on 20,000 fresh prompts. At $d = 5$, the demonstration-only learning curve is evaluated at every integer from 0 to 20. We compute ESS at the first crossing of this curve with the description-conditioned error, as in Definition 3.1. Table 6.1 reports results at $n = 0$, including evaluation at reliabilities that differ from the training distribution.

Networks that differ only in their seed are tested on the same prompts. Each network is tested at its training reliability and at $p = 1$ and $p = 0.9$; the networks trained with the second and third seeds are also tested at $p = 0.99$. Implied trust equals the regression coefficient of the network's prediction on $m^\top x$, and it can exceed 1.

Comparing networks on the same prompts. Because networks that differ only in their seed share their test prompts, the standard deviations in Table 6.1 reflect training alone and leave out variation from prompts. We therefore compare each network with the Bayesian predictor on the same prompts. With no demonstrations and $c = 10$, network ESS is 14.3 ± 0.5 at $p = 1$, 9.3 ± 0.2 at $p = 0.99$, and 4.66 ± 0.01 at $p = 0.9$, compared with Bayesian values of 16.1, 9.6, and 4.63 on the same prompts. On 200,000 prompts, the Bayesian value at $p = 1$ is 15.9, close to the limiting expression $c + d + 1 = 16$ of Proposition 4.1. At $p = 0.99$, the 200,000-prompt value, 8.96, lies below the networks' mean ESS, whereas the value on their own prompts, 9.6, lies above it. A network's ESS can exceed the Bayesian ESS on the same prompts, as it does at $p = 0.9$, because it uses the network's own demonstrations-only curve rather than the Bayesian one. For $c = 1$, network ESS is within 0.1 of the Bayesian ESS at every reliability; at $p = 1$, for example, it is 5.85 ± 0.03, compared with 5.89.

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). We train with Adam at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which is also optimal under squared error; evaluation uses the

Table 6.1: **Transformer ESS and implied trust under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. "Network" reports mean ESS across the indicated training seeds, with standard deviation; for a single seed, the uncertainty is the standard error over prompts. The train-matched Bayesian predictor uses the training reliability, and the calibrated predictor uses the test reliability. We show a single reference value when they coincide. Calibrated values, and coincident train-matched values, use 200,000 prompts, with standard errors at most 0.1. Other train-matched values use the network's test prompts; sampling variation can change these estimates by up to 0.6, so an estimated train-matched value can exceed its calibrated counterpart. "Trust" is implied trust averaged across seeds.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Train-matched	Calibrated	Trust
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

predicted mean and the loss in Equation (3.1). We use the optimizer of Garg et al. (2022), but unlike their model, ours uses bidirectional attention and no positional encoding. We choose $d = 5$ to keep training short.

Each demonstration token contains an input and its label. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description

is absent, its token is masked out of attention. The prediction is read from the query token. Labels and description values are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments below, which use 0 to 15.

Without a description, every network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$. Averaged over the demonstration counts used in training, its error is within 0.3 of the Bayesian reference.

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-demonstration ESS values are within 0.03 of the Bayesian values: 2.64 versus 2.66, 2.16 versus 2.15, 7.32 versus 7.35, and 4.38 versus 4.37, respectively.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with the specific description and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. The network ESS is 17.8, compared with a Bayesian value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the network captures most of the description's value but falls short of the reference. The standard deviation across seeds is 0.3, and the shortfall, 3.0 demonstrations, is larger than the 1.8 at $d = 5$.

We also train networks with a partial description, as defined in Section 4.2. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this $c = 10$ description, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayesian values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 4.2. The large-sample limit of that proposition is $k(c - a_0^{-1})/d = 5.94$, and the standard deviation across seeds at $n = 0$ is 0.03.

ESS with demonstrations. Figure 6.2 evaluates each network at its training reliability. For the less precise description, network ESS remains within 0.6 of the Bayesian value at every demonstration count whose crossing lies within the evaluation grid. For the more precise description, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. At $n = 5$, the three networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77,

compared with the Bayesian value of 5.20.

Reliability shift without demonstrations. For the $c = 10$ description, the implied trust of the networks trained at $p = 0.9$ is 0.87 to 0.89 across the three seeds. Their ESS on reliable descriptions is 9.5 ± 0.4, whereas the calibrated predictor reaches 15.9 and the train-matched predictor 10.8. The networks trained at $p = 1$ or $p = 0.99$ have implied trust between 0.95 and 1.02, so the two training reliabilities cannot be distinguished without demonstrations. At test reliability $p = 0.9$, full trust costs the networks trained at $p = 1$ little at $n = 0$: their ESS is 4.58 ± 0.03, compared with 4.70 for the calibrated predictor.

Reliability shift after demonstrations are observed. For the specific description, networks trained at $p = 1$ and $p = 0.99$ both behave approximately as if the description were certainly relevant when $n = 0$. Their behavior separates once demonstrations are available. At test reliability $p = 0.9$, the description reduces error for the networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward, as reported earlier in this chapter.

For the networks trained at $p = 0.99$, the reduction in error is 3.5 to 3.7 with six demonstrations and 0.08 to 0.13 with 20. The fully trusting Bayesian predictor behaves like the networks trained at $p = 1$. With eight demonstrations, for example, the first of these networks has error 4.6 with the description and 2.3 without it.

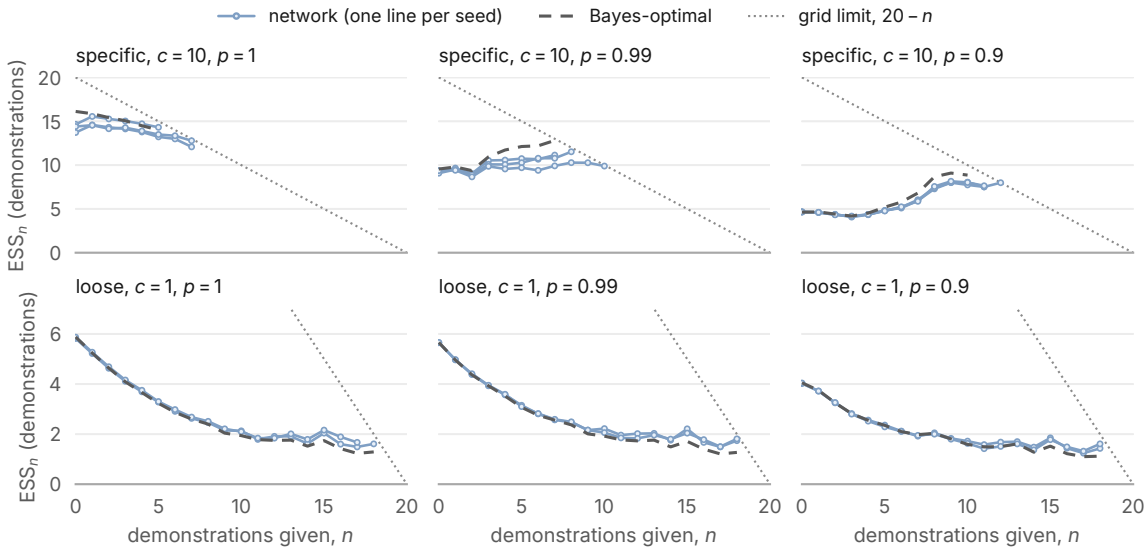


Figure 6.2: **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayesian reference is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

Chapter 7

Pretrained Language Models

The meta-trained networks learn description statistics from repeated tasks. We next ask whether pretrained language models use the same information when it is stated explicitly. We use numerical mean estimation, a scalar version of the regression problem with fixed input $x = 1$, following controlled tests of stated priors and in-context evidence (Gupta et al., 2025; Akata et al., 2026), and then a semantic classification task.

7.1 Task and Evaluation

Each task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$ and demonstrations drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, then rounded to integers. A description supplies a value m and uncertainty τ, for example: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." The underlying Gaussian construction has $w \mid m \sim \mathcal{N}(m, \tau^2)$. We express precision in demonstration units as $c = \sigma^2/\tau^2$: stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36.

We vary the stated precision, and add conditions stating a 50% probability that the description is wrong. In these conditions the description remains relevant; the intervention tests how models respond to the stated probability. We also hold m and the demonstrations fixed while describing a value generated at $c = 36$ as having $c = 4$. This isolates sensitivity to the stated uncertainty from sensitivity to the numerical value itself. Section 7.9 includes genuinely irrelevant descriptions and evaluations under actual mixtures of relevant and irrelevant descriptions.

The main comparison uses the same 200 tasks for five models with reasoning enabled and disabled: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol, and Claude Opus 4.6. Claude Haiku 5.5 is evaluated with reasoning at medium effort. Checkpoints are held fixed within model, although serving environments and decoding settings differ for some comparisons. Sections 7.5–7.10 report these details, response extraction, token limits, reruns, and a larger non-reasoning evaluation on 2,000 tasks.

A reference based on the prompt. The prompt specifies the observation noise and description, but not the task generator's baseline prior. Our *prompt-based Bayesian reference* therefore uses the stated description and a uniform fallback over the three-digit range. For a description asserted to be relevant, ignoring negligible boundary effects, its prediction is

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{7.1}$$

where $\bar{y}$ is the demonstration mean. Under the continuous, correctly specified Gaussian model with a flat baseline prior, the description has exact ESS c. There is no random-design dimension term because every observation measures the same scalar mean.

Intuitively, the description counts as c further demonstrations at m: it adds a precision of $1/\tau^2 = c/\sigma^2$, equal to that of c demonstrations. This is the conjugate normal model of Chapter 2 with $n_0 = c$, and the squared error is $\sigma^2/(c+n)$ with the description and σ^2/n without it.

This reference follows the prompt even when a stated precision or reliability differs from the data generator. It is therefore a model of inference from the stated information, rather than an oracle for every evaluation condition. When the description may be wrong, its posterior also depends on the chosen fallback; we report sensitivity to that choice.

For each model, we compute ESS from its own demonstrations-only curve and summarize it by the median over $n \in \{1, 2, 4\}$. The same calculation gives reference ESS 34.6 for $c = 36$, rather than the idealized value 36, because of finite-sample error and interpolation. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m measures reliance on the description; the reference coefficient is $c/(c+n)$ when the description is asserted to be relevant.

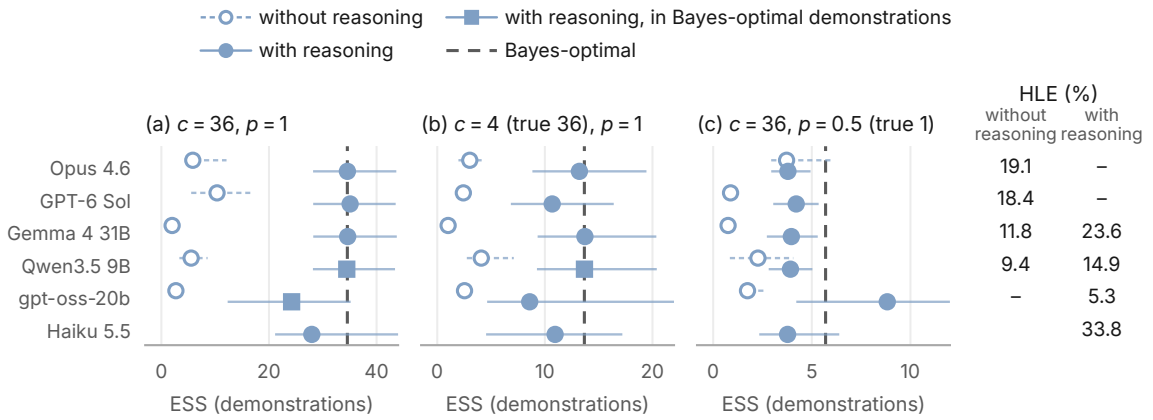


Figure 7.1: **Reasoning improves the use of stated precision, with model-specific exceptions.** Summary ESS on 200 tasks at $\sigma = 60$: median over $n = 1, 2, 4$, with paired 95% bootstrap intervals over tasks. Open and filled points denote reasoning disabled and enabled; dashed and solid intervals distinguish the modes. Vertical dashed lines show the prompt-based Bayesian reference. (a) Relevant description with $c = 36$: reference ESS 34.6. (b) The same value, generated at $c = 36$, is stated to have $c = 4$: reference ESS 13.7, because the value is more accurate than stated. (c) The description in (a) is assigned a 50% probability of being wrong, while remaining relevant: reference ESS 5.7 under the uniform fallback, which the prompt does not state. For GPT-6 Sol and Haiku 5.5, stating it changes which rule they follow and raises their ESS by at most 1.2 (Section 7.11). Square reasoning markers for Qwen3.5 9B in (a,b) and gpt-oss-20b in (a) use the reference's demonstrations-only curve because the models' own curves do not reach the crossing; these values are in reference-demonstration units. Haiku 5.5 uses medium effort. Intervals are clipped to the displayed range. HLE scores (Phan et al., 2025), as reported by Artificial Analysis (2026), provide descriptive context; dashes indicate unavailable scores for the evaluated mode, and blanks indicate modes not evaluated.

7.2 Reasoning and Stated Precision

Weak sensitivity without reasoning. For the $c = 36$ description, non-reasoning ESS ranges from 2.0 to 10.3 across the five models, compared with 34.6 for the reference (Figure 7.1a). Models generally underweight this precise description, and most respond weakly when its stated precision changes. At $n = 4$, GPT-6 Sol and Opus 4.6 assign weights 0.47 and 0.40 at stated $c = 36$, and 0.17 and 0.20 at stated $c = 4$, compared with reference weights 0.90 and 0.50. When the same value is described as $c = 4$ instead of $c = 36$, reference ESS falls from 34.6 to 13.7. GPT-6 Sol's falls from 10.3 to 2.4, and no other model's changes by more than 2.8 (Figure 7.1b).

Reasoning recovers precision weighting in several models. With reasoning enabled, Opus 4.6 and Gemma 4 31B reach approximately the reference ESS in both precision conditions; GPT-6 Sol and Haiku 5.5 at medium effort come close. Among diagnostically separated tasks at $n = 4$, the weight implied by an individual reply is within 0.05 of the reference weight in 89–100% of these models' replies (Section 7.11). This supports a change in how the description and demonstrations are combined, beyond an improvement in aggregate error alone.

Qwen3.5 9B and gpt-oss-20b also approach the reference precision weights, but arithmetic errors complicate their ESS. Their demonstrations-only curves can rise at larger n, preventing a crossing within the evaluated grid. For these cases, square markers in Figure 7.1 report sample equivalence using the reference's demonstrations-only curve, a different quantity from learner-specific ESS. Errors with the description also reduce gpt-oss-20b's ESS in some conditions. Reported HLE scores do not order these behaviors: some higher-scoring non-reasoning models use precision less accurately than lower-scoring reasoning models. This comparison is descriptive and does not isolate the mechanism responsible for the difference.

7.3 Stated Reliability and Wording

Reasoning models update relevance when the prompt says what a wrong description implies. Adding a stated 50% probability of error lowers reference ESS from 34.6 to 5.7 under the uniform fallback. Without reasoning, GPT-6 Sol's ESS falls from 10.3 to 0.9, more than the reference's, while no other model's falls by

more than a factor of 2.7, compared with a factor of about six for the reference. With reasoning, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B closely track a different rule: they retain the stated relevance probability and weight m by $p\,c/(c+n)$ (Table 7.7). This averages the relevant and irrelevant predictions using the *prior* mixture weights.

Bayesian inference instead updates relevance by comparing how likely the demonstrations are if the description applies and if it does not. The prompt says only that the mean is then "unknown", which leaves the second likelihood, and so the update, undefined; keeping the stated probability is one defensible response. We therefore repeat the condition for GPT-6 Sol at low effort and Haiku 5.5 at medium effort, changing only the warning to say that the mean "was instead drawn uniformly at random from the whole numbers 100 to 999", the fallback the reference assumes. On tasks where the reference and fixed-rule predictions differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5 (Section 7.11). Once the alternative is specified, both models update relevance from the demonstrations. The update is not always calibrated: with a stated 10% chance, Haiku 5.5 relies on the description less than the reference does, and its ESS falls to 17.9, against 27.5; GPT-6 Sol's is 25.0.

More explicit statistical descriptions are used more consistently. We test three alternative wordings with GPT-6 Sol and Haiku 5.5. Both closely follow Equation (7.1) under the original wording and "An earlier estimate put its mean at m, with a standard error of τ." Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$," they often return m and disregard demonstrations. With the first informal wording, GPT-6 Sol skips reasoning in 24–51% of replies per condition and usually returns m in those replies. Haiku 5.5 returns m in 71% of the replies for which it does reason.

These alternatives preserve the numerical values but do not specify an equally precise statistical interpretation. The statistical wordings explain how the uncertainty should enter the task; informal uncertainty expressions leave that relationship less explicit. As with the unstated fallback, the finding therefore concerns the specificity of the description as a whole, including its instructions for interpreting uncertainty, rather than invariance to semantically equivalent phrasing.

7.4 A Semantic Task

Reasoning models use semantic descriptions almost as the reference does. In our semantic task, a company files messages into one of two folders by a hidden policy that requires two of six properties, written in ordinary language, to be present or absent. The prompt states that each of the 60 such policies is equally likely, so the Bayesian reference is exact (Section 7.13). A specific note states one requirement and names the other property. With reasoning, Haiku 5.5 and GPT-6 Sol value it at 9.9 and 9.5 demonstrations at $n = 2$, against 11.4 for the reference. Given the same note with a stated 50% chance of being wrong and the fallback stated, they update it from the demonstrations. Without reasoning, GPT-6 Sol makes poor use of the note.

7.5 Additional Language-Model Results and Evaluation Details

We generate 2,000 tasks. Six open-weight models answer all tasks without reasoning; we refer to these experiments as the *full run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Sections 7.1–7.3, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The experiments in Sections 7.1–7.3 use the first 200 tasks. Unless stated otherwise, the results in this section and those that follow use the full run without reasoning.

Table 7.1 reports summary ESS, defined as the median over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 7.2 shows the underlying learning curves. Tables A.1 and A.2, in Appendix A, report ESS at every evaluated demonstration count; Table 7.2 compares interpolation methods; Table A.3, also in Appendix A, reports description weights for the models with a distribution readout; and Table 7.3 evaluates genuinely unreliable descriptions.

Throughout these sections, the *prompt-based Bayesian reference* uses the probabilistic claims in the prompt together with an explicit assumption for the unspecified fallback prior. It is optimal under that assumed model. When the stated description differs from the data-generating process, it need not minimize error under the evaluation distribution. The reference labeled "Bayes-optimal" in the plots and tables has

this meaning.

7.6 Prompts, Tasks, and Model Readouts

Model inclusion. We retain a model only if it produces a number in at least three quarters of replies overall and in at least half of replies without a description at each of $n = 1$, 2, and 4. The latter conditions determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction. For example, OLMo 2 7B's summary ESS for the $c = 36$ description is 21, but this value comes from a readout that rests on a minority of the model's probability.

The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution over the 900 three-digit number tokens. We call this the *distribution readout*. These three models encode every three-digit integer as a single token, and we renormalize the first-token probabilities over these tokens. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Tables 7.1 and 7.4). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
```

Table 7.1: **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals. An upper bound, $\leq$, indicates that one demonstration alone outperforms the description-conditioned prediction. Right: the coefficient on the stated value at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1$, 2, 4						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

```
548
431

Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun of Section 7.3 that states the fallback, its last clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table A.1).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 7.4, A.1, A.2, and A.3 report this quantity as "number". Tables 7.4 and A.3 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout extracts the first number with three digits before an optional decimal part and retains the decimals. Without reasoning, decoding is greedy with a limit of eight new tokens. Those predictions were recorded using an earlier readout that retained only the integer part; this rarely changes the value because decimal outputs are uncommon. With reasoning, extraction is applied to the final answer. Replies without a qualifying number are excluded from both error curves and weight regressions.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 63% of written replies across all conditions, so Table 7.1 reports only its distribution readout. In the wording experiment, however, its numeric response rate exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

7.7 ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 3.1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. The ESS at n is the first interpolated demonstration count t that matches the description-conditioned error, minus n.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$,

we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we report $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. Summary ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error ranges from 9 to 23, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below.

Reference prior and exact sample equivalence. The prompt-based Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the demonstration mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description, the idealized continuous model with a flat baseline prior has exact sample equivalence c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as a prior probability of irrelevance and updates that probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Sections 7.1–7.3, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and

35.64. This is the scalar analogue of $(1 - r)c$ in Chapter 4, with $r = \tau^2/100^2$. We use the prompt-based reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Chapters 4–5, a reference that accepts an incorrect description can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c+n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 7.2 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using pool adjacent violators; and a relative-least-squares fit of $A/(n + n_0) + B$ to the baseline errors. The final form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the estimator in Table A.1, but 9.2 under the fitted curve.

7.8 Results of the Full Run

Summary ESS. Across the six models, the $c = 36$ description has summary ESS 1.4 to 7.6, compared with reference ESS 36 on these tasks, and ESS at most 15 at any of $n = 1$, 2, and 4 (Table 7.1). The $c = 1$ and $c = 4$ descriptions have summary ESS 0.0 to 1.9 and 0.2 to 4.1, compared with 1.3 and 4.4. ESS rises with c mainly because a more specific description has a value closer to the task, as Table 7.5 shows.

Description weights. At $\sigma = 60$, with no demonstrations and no stated reliability, the models return the description value (weight 0.92 to 1.10), although Phi-4 and

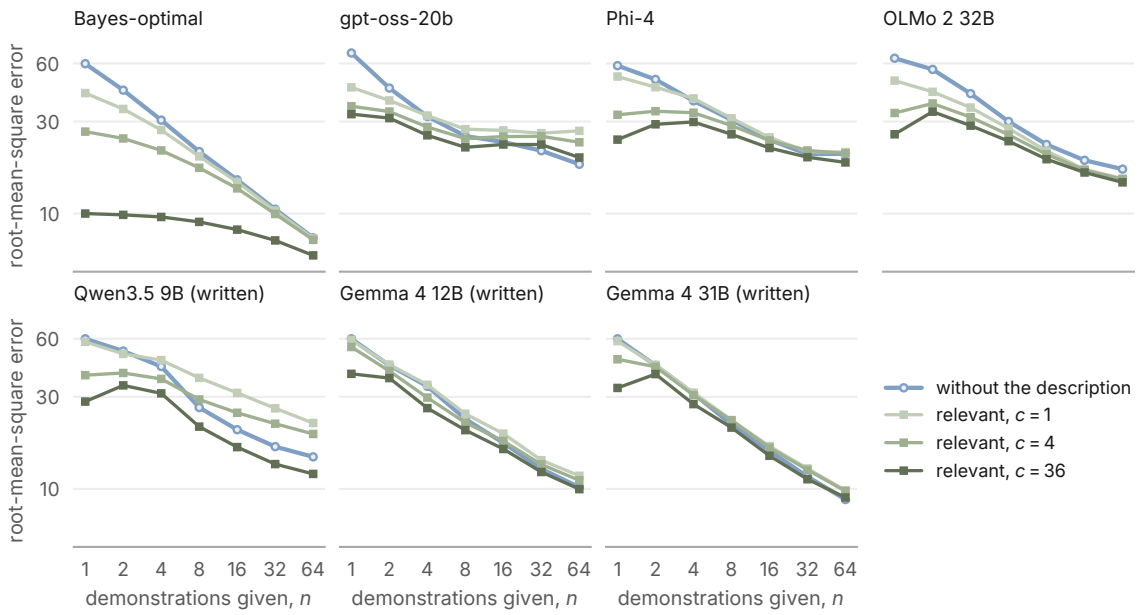


Figure 7.2: **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table A.3).

Table 7.2: **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Bayes-optimal	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

OLMo 2 32B then begin only 20 to 23% and 50 to 59% of their replies with a number. At $n = 1$, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything falls, whereas the reference weight rises from 0.50 to 0.80 to 0.97 (Table 7.1). At every n from 1 to 64 and at both noise levels, the spread of these three models' weights across c is at most 0.16, compared with a reference spread of 0.34 to 0.71 (Table A.3). The weights of the two Gemma models do rise with c, by about 0.25 between $c = 1$ and $c = 36$ at $n = 1$, half the reference rise. The weight of Qwen3.5 9B does not. Models therefore underweight a precise description and, with many demonstrations, overweight the least precise one. At $n = 64$, gpt-oss-20b still assigns weight 0.23 to the $c = 1$ value, compared with reference weight 0.02, and that description raises its error (ESS -57).

Stated unreliability. A stated 10% or 50% chance of being wrong leaves the ESS of gpt-oss-20b and OLMo 2 32B about where it was, and Phi-4's ESS even rises, from 7.6 to 8.7 and 9.7. Qwen3.5 9B and Gemma 4 31B have lower ESS, but by about as much for 50% as for 10%. Gemma 4 12B's ESS falls at 10% and rises at 50% (Table 7.1). No model's ESS falls in proportion to the stated probability, whereas reference ESS falls to 22 and 5.8.

7.9 Unreliability, Description Accuracy, and Wording

Irrelevant descriptions. An irrelevant description uses the value from a different, independently sampled task. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference. Also at $n = 4$, their weight on an irrelevant $c = 36$ value is 0.01 to 0.33, compared with 0.18 to 0.44 on a relevant one.

Actual unreliability. The stated-unreliability experiments keep descriptions relevant and vary what models are told. Table 7.3 instead evaluates the mixture in Chapter 3. We combine squared errors from the relevant and irrelevant conditions, on the same tasks, with weights p and $1 - p$, and then estimate ESS. We evaluate models both with and without disclosure of the irrelevance probability. The fully trusting

reference always treats the description as relevant; the calibrated reference starts from irrelevance probability $1 - p$ and updates it from the observations.

At actual reliability $p = 0.9$, a $c = 36$ description has ESS 5.5 for the calibrated reference and 0.2 for the fully trusting reference. Model ESS ranges from 0.4 to 3.2 when the reliability is not stated and from -0.1 to 4.0 when it is stated. Model weights depend weakly on p, and stating the probability changes ESS by at most two demonstrations in either direction. Their nearly fixed weight on the description thus acts as a hedge between full trust and calibration. At $p = 0.5$, the description is worth less than one demonstration to the calibrated reference and at most one to every model.

Table 7.3: **Description ESS under actual unreliability.** Relevant and irrelevant $c = 36$ conditions are combined with weights p and $1 - p$. Results are summary ESS with 95% task-bootstrap intervals for the full run without reasoning at $\sigma = 60$. The fully trusting reference assumes relevance; the calibrated reference uses the actual reliability and updates it from demonstrations. A bound indicates that one demonstration alone outperforms the description-conditioned prediction at one or more summary demonstration counts.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Fully trusting	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Calibrated	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	-0.1 [-0.1, 0.0]	-0.1 [-0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	-1.0 [-1.0, -0.7]	-0.1 [-0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

Wording without reasoning. We repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 7.4 uses the full-run

tasks and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. A coefficient above 1 means that predictions vary more strongly with the stated value than the value itself. Nevertheless, none of the tested wordings yields the reference's sensitivity to specificity: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison also varies how precisely the task is specified.

Separating stated specificity from description accuracy. For each task, we take a description value generated either at $c = 36$ or at $c = 1$ and present it with each stated specificity in $\{1, 4, 36\}$. The tasks and demonstrations match the full run, and the experiment includes its own no-description prompts at every n (Table 7.5). Repeated conditions reproduce the full run exactly except for gpt-oss-20b. Its distribution readout changes by a median of 0.8 to 1.2 across conditions, with a maximum difference of 39; its written replies change in 12% of cases, with median difference 0. A newer library version is a possible explanation.

The prompt-based reference accepts the stated specificity. A value generated at $c = 1$ but stated as $c = 36$ therefore has reference ESS of about -0.8; a model that discounts the stated specificity can outperform it in that misspecified condition. For models without reasoning, description accuracy primarily determines ESS. When the value is generated at $c = 1$, each model's ESS remains between -0.1 and 2.0 across all stated specificities. When the value is generated at $c = 36$, each model's ESS is higher, but from stated $c = 1$ to stated $c = 36$ it rises by at most 1.2, whereas reference ESS rises from 2.5 to 36. For example, gpt-oss-20b has ESS 2.7, 2.6, and 2.8 at stated $c = 1$, 4, and 36. Phi-4's ESS even falls, from 12 at stated $c = 1$ to 7.6 at stated $c = 36$, the reverse of the reference order.

7.10 Reasoning Runs and Execution Settings

Shared evaluation prompts. Every model in Figure 7.1, as well as Olmo 3 7B Think (Team Olmo, 2025), is evaluated on the same 7,800 prompts from the first 200

Table 7.4: **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate; italics mark rates below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated specificity remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Table 7.5: **Sensitivity to stated specificity with the value and demonstrations held fixed.** Results use $\sigma = 60$, the full-run tasks, and the readout conventions of Table 7.1. For a value generated at $c = 36$, the table reports its weight at $n = 1$ and summary ESS under each stated c. Results for the 200-task runs of Sections 7.1–7.3 appear in Table 7.6.

	weight, $n = 1$			ESS		
stated c	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.43	0.49	0.72	0.2	0.3	1.4

tasks. These include no-description prompts at $n = 1$, 2, 4, ..., 64, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b run omits the two additional points and contains 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. The two modes receive the same prompts, although their serving environments can differ as detailed below. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average, and at least 97% of replies reach it in every cell. An earlier run with a limit of 384 tokens finished only 28 to 54% of the replies in the stated-reliability conditions, and the difference between a stated 10% and 50% that this run seemed to show came from which replies finished. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 loop to the token limit, a failure its model card warns of. Each is rerun once using the model card's recommended reasoning sampling settings (temperature 1.0, top-p 0.95, top-k

20, presence penalty 1.5); 10 remain unfinished.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun using the model card's recommended sampling settings (temperature 1.0, top-p 0.95, top-k 64).

Olmo 3 7B Think runs in vLLM with greedy decoding and at most 32,768 new tokens, the limit its model card recommends; at 8,192 tokens, only 12 of 20 pilot replies finished. We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model uses 9,215 reasoning tokens on average. Its long greedy outputs are sensitive to hardware: an earlier stated-specificity run on a different GPU produces the same final number on only half the prompts. We therefore report one run. Despite its extensive reasoning, this model learns little from the demonstrations; its prediction behavior is reported below.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`) runs through the Codex command-line agent under a subscription, in a read-only sandbox, with an empty configuration directory, so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate reasoning disabled, with no reasoning tokens, and reasoning enabled at low effort. The agent's own instructions surround the prompt, much as a provider's system prompt surrounds a prompt sent through an API. An earlier configuration loaded user instructions and allowed shell commands, which the model used in about 8% of replies, mostly to average long demonstration sequences. Without reasoning, that configuration also gave the description much less weight (ESS 1.5 rather than 10.3 at $c = 36$); with reasoning, every summary ESS changes by at most 2.7. The wording runs below used the earlier configuration, and the model invoked a shell in 4 of their 5,400 replies.

Claude Opus 4.6 and Haiku 5.5 run through the Claude Code command line under a subscription, with one prompt per call, no tools, no project or user instructions, and a one-line system prompt. Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. When enabled, it reasons before at least 99% of replies in each description condition and averages 599 reasoning tokens across all replies. Without a description, it reasons before only 1%, 24%, and 79% of replies at $n = 1$, 2, and 4, respectively. Haiku 5.5

(`claude-haiku-5-5`) is evaluated at medium and low effort, using the command line's default sampling temperature. At medium effort, it reasons before 37 to 88% of replies at $n = 1$ across conditions and before nearly all replies from $n = 2$ onward, with 270 reasoning tokens on average. Its no-description error stays close to the reference through $n = 48$; at $n = 16$, for example, it is 15.8 against 14.8, with two replies far off. Its error rises at $n = 64$, where it divides the sum by 60 instead of 64 in 9 of 200 replies.

Interpreting the comparison between modes. The experiments hold the model checkpoint and prompts fixed within each paired comparison, but reasoning settings, serving software, readout, retry policy, and access to calculation tools are not uniform across all runs. The results describe the behavior of the evaluated configurations. They do not by themselves isolate a single internal mechanism responsible for the difference between modes.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B assign approximately the reference weight to the description: at $n = 4$ and $c = 36$, their weights are 0.89 and 0.90, compared with 0.90 for the reference. However, both make errors when averaging longer demonstration sequences. Qwen3.5 9B has no-description root-mean-square error 15.1 at $n = 16$ and 35.4 at $n = 32$, compared with reference errors 14.8 and 10.0. From $n = 32$ onward, between 4 and 24 replies per cell differ from the sample mean by more than 20; one reply gives 945 for numbers averaging 467.

For these models, the baseline curve rises after 16 demonstrations, and some description-conditioned errors never meet it within the evaluated grid. Table 7.6 therefore also reports, in parentheses, a comparison using the Bayesian reference's baseline curve. This quantity is the number of demonstrations required by the reference without a description to match the model's error, rather than ESS in the model's own units. The gpt-oss-20b model also makes arithmetic errors with descriptions, reducing the reported ESS in those conditions; for example, it obtains 1.3 when the prompt states a 10% chance of being wrong. About 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the two variances. On the reference's baseline, its $c = 36$ description is therefore worth 24 demonstrations, well below the reference's 34.6, although its weight is right. A fitted weight hides such

errors, whereas ESS does not.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks, and using either baseline changes their ESS by at most 1. At low effort, Haiku 5.5 often skips reasoning and repeats the demonstration. Its description weight at $n = 1$ is consequently low even in conditions whose ESS is close to the reference value.

Table 7.6: **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS is computed using each run's baseline curve; parentheses show the value using the Bayesian reference's baseline where it differs. "10% wrong" and "50% wrong" denote the stated probability for a $c = 36$ description. The last column uses a value generated at $c = 36$ but described as $c = 4$. "Reasoning" identifies the mode and provider effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator. Olmo 3 7B Think's no-description error never drops below its error at one demonstration, so its reported crossings require particular caution; using the reference baseline gives a different comparison rather than resolving the weak model baseline.

Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10% wrong	50% wrong	Stated $c = 4$ True $c = 36$
Bayes-optimal	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.7	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	> 60 (21)	3.9	> 60 (14)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	11 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.4 (3.3)	26 (24)	≤ -1.0	27 (26)	5.7 (5.6)	16 (8.1)
Olmo 3 7B Think, with reasoning	≤ -0.7 (≤ -0.9)	-0.7 (-0.9)	> 60 (≤ -0.5)	≤ -1.0	> 60 (0.8)	> 60 (0.8)	> 60 (-0.7)

7.11 Prediction Rules and Reliability Updating

Rules inferred from individual replies. A weight fitted across tasks can mix replies that follow different rules. At $n = 4$, we estimate the description weight implied by an individual prediction as $(\hat{y} - \bar{y})/(m - \bar{y})$. We use the 61 tasks per cell where $|m - \bar{y}| \geq 30$ and exclude replies without a number. We classify a reply as consistent with a rule if its implied weight is within 0.05 of that rule's weight. The

candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible replies. Without reasoning, 79% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think instead repeats the description value in 96 to 97% of replies at $n = 1$, regardless of stated specificity. At $n = 4$, 70 to 77% of its replies match none of the rules, and only 3 to 8% match the reference. From task to task, it predicts the description value, the demonstration mean, or an invented pattern, and it makes arithmetic errors. Its no-description root-mean-square error remains near 55 from $n = 1$ to 32, compared with reference error 10 at $n = 32$. Reasoning mode is therefore not sufficient for the desired behavior in every tested model.

Within each cell, Haiku 5.5's replies split by whether it reasons. At $n = 1$, replies with reasoning assign weights 0.50, 0.80, and 0.97 to the relevant $c = 1$, 4, and 36 values, the reference weights to two decimals, at either effort. Replies without reasoning assign weight 0.00 to 0.05 and repeat the demonstration. Its overall weight at $n = 1$ mixes the two kinds of reply: 0.44, 0.66, and 0.74 at medium effort. On these three descriptions, it reasons before 34 to 53% of replies at low effort, against 72 to 82% at medium effort, so its overall weight at $n = 1$ is lower at low effort and nearly flat, at 0.36, 0.34, and 0.41. With more demonstrations it reasons more often, and its weight approaches the reference again: at $n = 4$ and low effort, it is 0.17, 0.48, and 0.86, compared with 0.20, 0.50, and 0.90. The choice to reason is the model's own, so this split describes its replies rather than the result of an experiment.

A fixed reliability rule. Table 7.7 reports description weights for $c = 36$ when the prompt states a probability of being wrong. A predictor that combines the description-conditioned and no-description predictions with fixed weights p and $1 - p$ assigns weight $p\,36/(36 + n)$ to m. Under the original warning, which leaves the fallback unstated, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to this fixed rule. Opus 4.6 and Haiku 5.5 at medium effort fall below the reference weight by $n = 4$. Haiku 5.5 at low effort, which often skips reasoning at $n = 1$, comes

closest to the reference weight. gpt-oss-20b assigns nearly the weight of a reliable description whatever the stated probability. These are behavioral comparisons; they do not identify the internal procedure used in every reply.

Sensitivity to the unspecified fallback prior. The prompt states only that, if the description is wrong, "the mean is unknown". A Bayesian relevance update compares the likelihood of the observations under the description with their likelihood under this fallback. Because the prompt leaves the fallback distribution unspecified, it does not determine a unique Bayesian relevance update.

For the $c = 36$ description with a stated 50% chance of being wrong, the reference's fitted weights on the first 200 tasks at $n = 1$, 2, 4, 8, and 16 are 0.58, 0.61, 0.64, 0.58, and 0.54 with the uniform fallback over 100 to 999. They are 0.49, 0.49, 0.53, 0.49, and 0.45 with the actual task prior $\mathcal{N}(500, 100^2)$; 0.52, 0.54, 0.58, 0.53, and 0.49 with $\mathcal{N}(500, 200^2)$; and 0.72, 0.74, 0.74, 0.67, and 0.60 with a uniform fallback over 0 to 2,000. The fixed rule gives 0.49, 0.47, 0.45, 0.41, and 0.35.

By four demonstrations, the reference weight exceeds the fixed-rule weight for every fallback examined. Thus the observed agreement of several models with the fixed rule persists across these choices of reference. The size of their departure from Bayesian updating depends on the fallback, however; under the actual task prior, the reference update is small before four demonstrations.

Stating the fallback. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, for $c = 36$, stated chances of 10% and 50%, and $n = 1$, 2, 4, with the fallback stated as the reference's uniform over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each. Table 7.8 compares their description weights with those of the original runs. With a stated 50% chance, the reference and the fixed rule differ by at least 5 on 123 to 155 tasks per cell. On those tasks, the share of replies closer to the reference than to the fixed rule rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5. With a stated 10% chance, at most 7 tasks per cell separate the two rules, so this share is uninformative there.

Summary ESS, computed against each model's curve without the description from the main run, changes little at 50%: from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5, against 5.7 for the reference, because the two rules give similar errors

Table 7.7: **Description weights under stated unreliability.** Coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. "Fixed" uses weight $p\,36/(36 + n)$; the reference updates relevance using the uniform fallback over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition, and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. Several reasoning configurations closely track the fixed rule.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.50	0.46	0.43	0.88	0.84	0.74
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

at $n \leq 4$. At 10%, it falls from 29.7 to 25.0 and from 32.6 to 17.9, against 27.5: with the fallback stated, Haiku 5.5 discounts the description more than the reference does, and GPT-6 Sol slightly more.

7.12 Wording with Reasoning Enabled

GPT-6 Sol. We evaluate the three alternative wordings at low effort for the three relevant descriptions and $n = 1$, 2, and 4, using the first 200 tasks. Of 5,400 replies, 5,399 contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before every reply, and its description weight is within 0.01 of the reference in every cell, as under the original wording.

Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 24 to 51% and 11 to 50% of replies per cell, respectively. Of those replies, 83% and 87% repeat the description value. At $c = 1$ and $n = 1$, the fitted description weights are consequently 0.89 and 1.00, compared with reference weight 0.50. Among replies that do include reasoning, 68% and 43% are within 1 of the reference prediction, while 36% and 68% repeat the description value. These categories overlap at $c = 36$, where

Table 7.8: **Stating the fallback changes the reliability rule.** Coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 7.7. "Unstated": the original warning, "the mean is unknown"; "stated": the mean is otherwise uniform over 100 to 999, as the reference assumes.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

the description value and reference prediction can be close. Under any reading of τ, the reference prediction moves with the demonstrations, yet most replies that skip reasoning repeat the description value.

Haiku 5.5. We evaluate the same alternatives at medium effort; all 5,400 replies contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before at least 72% of replies in each cell, and 99% of the replies with reasoning lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

These wording changes affect the precision of the probabilistic specification as well as the model's behavior. All provide a value m and a scale τ, but "give or take" and "$\pm$" do not uniquely identify τ as a prior standard deviation or establish a sampling model. The experiment therefore shows that accurate combination of descriptions and demonstrations under explicit statistical statements does not automatically transfer to these less explicit formulations. It does not isolate sensitivity to semantically equivalent descriptions.

7.13 Details of the Semantic Task

Task. A message has six binary properties: the sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property one way or the other, in one of three phrasings per value, such as "I'm a contractor from another company." or "I'm on the sales team here." A policy uses exactly two properties and requires each to be present or absent; a message goes into folder A when both requirements hold, and into B otherwise. The prompt describes this family of policies and states that the company's policy was chosen at random, with each of the 60 possible policies equally likely. It then gives the policy note, n previously filed messages with their folders, and four new messages, and asks for the probability that each new message goes into A, as a number between 0 and 1. Without reasoning, GPT-6 Sol reads each property correctly in 97 to 100% of 96 messages, so the phrasings are unambiguous.

Policy notes. The broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," which leaves 20 policies. The specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," which leaves 2. The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." It is wrong in half the tasks, when it describes an independent policy.

Reference and ESS. The reference's probability of A is the share of policies consistent with the note and the filed messages that send the new message to A. For the unreliable note, it mixes these policies with those consistent with the filed messages alone, weighting the note by the stated 50% updated on the demonstrations. On 3,000 simulated tasks, it is calibrated in every condition, and its posterior probability that the unreliable note is right matches how often it is. The loss is the squared error of the probability against the true folder, averaged over the four messages. ESS follows Definition 3.1, against each learner's own curve without a note at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, interpolated linearly; on simulated tasks, this grid moves the reference's ESS by at most 0.6 from its value read at every n.

Runs. We use 200 tasks, each a policy draw, and 19 cells: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4; this gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort and GPT-6 Sol without reasoning and at low effort, through the command-line agents of Section 7.10, with no tools. No reply called a tool. Of the replies, 14 from GPT-6 Sol at low effort and one from Haiku 5.5 give no probability and are left out. Haiku 5.5 reasons before every reply, with a median of 3,008 reasoning tokens; GPT-6 Sol at low effort reasons before 91%, with a median of 142.

Results. Table 7.9 reports the ESS of each note. With reasoning, both models are within 2 of the reference in every cell but one, GPT-6 Sol's specific note at $n = 4$: 26.1 against 14.8. Its loss with that note, 0.008, is close to the reference's 0.007, but its own curve without a note flattens past $n = 16$, with loss 0.022 at $n = 20$ against 0.004, so it needs more of its own demonstrations to match the note. For the unreliable note, we compare predictions with the reference and with a fixed rule that keeps the stated 50%, on messages where the two differ by at least 0.05. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5, 77% and 87% for GPT-6 Sol at low effort, and 45% and 45% for GPT-6 Sol without reasoning. Without reasoning, GPT-6 Sol learns slowly from demonstrations: its loss without a note is still 0.078 at $n = 32$, against 0.000 for the reference. In its own unit, a note can therefore be worth as many demonstrations as to the reference or more, such as 13.5 for the specific note at $n = 2$, although its loss with that note is 0.130, against 0.027.

Table 7.9: **ESS of each policy note in the semantic task.** ESS on 200 tasks, against each learner's own curve without a note. "≤ 0": the note did not lower the loss; "> 28": the curve without a note does not reach the loss with the note by $n = 32$. Paired 95% bootstrap intervals over tasks for the specific note at $n = 2$: reference $[9.7, 12.9]$, Haiku 5.5 $[8.7, 11.5]$, GPT-6 Sol at low effort $[7.8, 12.7]$, and without reasoning $[3.7, 21.9]$.

Note	n	Reference	Haiku 5.5, medium	GPT-6 Sol, low	GPT-6 Sol, none
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Chapter 8

Discussion

Description ESS connects two questions: how much task information is available, and how much of it a learner uses. In our model, a reliable description can replace many demonstrations because it constrains the task across directions as well as increasing precision. Unreliability changes this relationship: without demonstrations, even an arbitrarily precise description has bounded value. Demonstrations can then serve a second purpose, providing evidence about whether the description applies. Sufficiently precise partial descriptions can create another form of complementarity, becoming more valuable as examples resolve unspecified aspects of the task.

The empirical results show that these benefits are learnable but not automatic. Meta-trained Transformers approach the Bayesian benchmark under matched task distributions, while their responses to reliability shifts reflect training. Pretrained language models make substantially better use of explicitly stated precision with reasoning enabled. With reasoning, they also update relevance from the demonstrations once the prompt says what a wrong description implies, and they use semantic descriptions of a rule almost as the reference does. When the prompt leaves this open, they keep the stated probability fixed, and when it states uncertainty informally, they often ignore the demonstrations. These findings characterize observable prediction behavior; they do not identify a unique internal mechanism or establish a general separation between understanding and computation.

8.1 Limitations

The theory assumes Gaussian linear regression and a particular mixture model of irrelevance. The language-model tasks are small and state their probabilistic information explicitly; they are not unrestricted instruction following. The main comparison uses 200 tasks, and the reasoning-enabled wording test covers two models. Some mode comparisons also differ in serving or decoding settings. The original reliability prompts leave the fallback prior unspecified; we report several choices, and the rerun that states it covers two models. Extending the framework to open-ended semantic constraints, structured tasks, and richer or heavy-tailed description priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these relationships extend.

The networks are trained only at $d = 5$ and, for the higher-dimension experiment, at $d = 10$. Because the language-model task is scalar, it has no random-design dimension term and no partial description, and its models are post-trained chat models. Models without reasoning are evaluated through either the distribution readout or the written readout, so comparisons across these models mix differences between models with differences between readouts.

8.2 Future Work

We outline four directions for future work. The first two follow from the limitations and findings above, the third from the descriptions studied here, and the fourth from the networks of Chapter 6. None of them is established in this thesis.

Partial descriptions and the dimension term in language models. A precise partial description can become more valuable as demonstrations resolve what it leaves unspecified (Proposition 4.2), whereas the ESS of a reliable complete description falls toward $c - a_0^{-1}$ (Corollary 5.2). At $d = 16$, $a_0 = 10$, and $c = 100$, ESS increases from 8.1 without demonstrations to 36 with 100 demonstrations for $k = 8$, and falls from 117 to 109 for $k = 16$ (Table 4.1). The networks of Chapter 6 capture this increase with $k = 3$ of $d = 5$ directions described, and their ESS at $c = 10$ is larger at $d = 10$ than at $d = 5$. The language-model task, however, has a single unknown mean. A task with several unknown quantities, of which the description states only some, would test whether a language model's ESS has a dimension term, and whether a partial

description becomes more valuable to it as demonstrations accumulate.

Updating reliability. With reasoning, GPT-6 Sol and Haiku 5.5 update relevance from the demonstrations once the prompt states the fallback, but the update is not always calibrated: with a stated 10% chance, Haiku 5.5 relies on the description less than the reference does, and its ESS falls to 17.9, against 27.5; GPT-6 Sol's is 25.0. This rerun covers two models, $c = 36$, and $n \leq 4$, and its descriptions remain relevant. Three questions remain open. Do the other reasoning models update relevance once the fallback is stated? Why does Haiku 5.5 discount the description more than the reference does at a small stated chance? And do reasoning models also update relevance under actual mixtures of relevant and irrelevant descriptions, which Table 7.3 evaluates only without reasoning?

Structural descriptions. Apart from the semantic task, the descriptions studied here state where the task lies, through a center m with a stated precision. Many instructions instead state what kind of task it is, for example that only three features matter, or that the answer is monotone in an input. Such a description narrows the hypothesis class rather than pointing at one task. One case of this kind is already covered. A description that confines w to a known s-dimensional subspace states the other $k = d - s$ directions exactly, so it is a partial description of those directions in the limit $c \to \infty$. A heuristic calculation suggests that such a description keeps its value as demonstrations accumulate. Under a flat baseline prior, it would lower the Bayes-optimal error from about d/n to about s/n, so its ESS would be about $n(d/s - 1)$, which grows with the number of demonstrations. Structures that do not fix any direction, such as a sparse task whose relevant features are unknown, or a monotone one, are not covered. The definition of ESS and the network and language-model tests apply to them as well, and the semantic task of Section 7.4 gives a first test of one in a language model: its broad note names a property the policy uses without stating its required value, and with reasoning, Haiku 5.5 and GPT-6 Sol value it at 1.3 demonstrations at $n = 2$, against 1.4 for the reference (Table 7.9). Moreover, a wrong structural description would leave a fully trusting predictor with a bias that no number of demonstrations removes.

Where a network stores its trust. The meta-trained networks inherit their trust from training (Chapter 6). A natural next step is to locate the components of a network that hold this trust, and to test by intervention whether changing them changes the network's implied trust without retraining. Such a study would connect the behavioral measure of this thesis to the mechanisms by which networks learn in context.

Bibliography

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Artificial Analysis. Evaluations. https://artificialanalysis.ai/evaluations, 2026. Accessed 31 October 2026.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Chatlatanagulchai, W., Thonglek, K., Reid, B., Kashiwa, Y., Leelaprute, P., Rungsawang, A., Manaskasemsak, B., and Iida, H. On the use of agentic coding manifests: an empirical study of Claude Code. In *International Conference on Product-Focused Software Process Improvement (PROFES)*, 2025.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley, second edition, 2006.

Dalal, S. R. and Hall, W. J. Approximating priors by mixtures of natural conjugate priors. *Journal of the Royal Statistical Society: Series B (Methodological)*, 45(2): 278–286, 1983.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1 (4):893–914, 2006.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*. CRC Press, third edition, 2013.

Gemma Team. Gemma 4. https://huggingface.co/google/gemma-4-31B-it, 2026. Model card.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

He, Z., Qiu, T., Shirado, H., and Sap, M. Martingale score: An unsupervised metric for Bayesian rationality in LLM reasoning. In *NeurIPS*, 2025.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help transformers learn linear models in-context. In *ICLR*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

O'Hagan, A. and Pericchi, L. Bayesian heavy-tailed models and conflict resolution: A review. *Brazilian Journal of Probability and Statistics*, 26(4):372–401, 2012.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Phan, L., Gatti, A., et al. Humanity's last exam. *arXiv preprint arXiv:2501.14249*, 2025.

Pichon, F., Dubois, D., and Denœux, T. Relevance and truthfulness in information correction and fusion. *International Journal of Approximate Reasoning*, 53(2): 159–175, 2012.

Qiu, L., Sha, F., Allen, K., Kim, Y., Linzen, T., and van Steenkiste, S. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17:1238, 2026.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Selby, D., Iwashita, Y., Spriestersbach, K., Saad, M., Bappert, D., Warrier, A., Mukherjee, S., Kise, K., and Vollmer, S. Had enough of experts? Quantitative knowledge retrieval from large language models. *Stat*, 14:e70054, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

Appendix A

Language-Model Results at Every Number of Demonstrations

This appendix collects the language-model tables of Chapter 7 at every number of demonstrations. Tables A.1 and A.2 report ESS at every evaluated demonstration count, and Table A.3 reports description weights for the models with a distribution readout. Tables A.1 and A.3 each continue onto a second page.

Table A.1: **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. An upper bound, $\leq$, marks error above the first baseline point; $> 64-n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60$, $c=1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	>0
	$\sigma=60$, $c=1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma=60$, $c=4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	>0
	$\sigma=60$, $c=4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma=60$, $c=36$, relevant		36	36	37	38	39	>32	>0
	$\sigma=60$, $c=36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma=30$, $c=1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	>0
	$\sigma=30$, $c=4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	>0
	$\sigma=30$, $c=36$, relevant		42	43	43	43	42	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	>0
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma=60$, $c=36$, relevant, 10% wrong		13	22	26	35	38	>32	>0
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma=60$, $c=4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	>0
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma=60$, $c=36$, relevant, 50% wrong		3.1	5.8	12	21	31	>32	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma=60$, $c=1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma=60$, $c=4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma=60$, $c=4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma=60$, $c=36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma=60$, $c=36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma=30$, $c=1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma=30$, $c=4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma=30$, $c=36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma=60$, $c=1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma=60$, $c=4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma=60$, $c=4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma=60$, $c=36$, relevant	0.86	15	7.6	4.7	5.6	8.5	>32	>0
	$\sigma=60$, $c=36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma=30$, $c=1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma=30$, $c=4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma=30$, $c=36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	>32	>0

	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.85 \leq 0.0 \leq -1.0$			-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.86 \leq 0.0 \leq -1.0$			-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.85 \leq 0.0 \leq -1.0$			-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.86 \leq 0.0 \leq -1.0$			-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	$0.89 \leq 0.0$		-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	$0.90 \leq 0.0$		-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	$0.90 \leq 0.0 \leq -1.0$			-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.96 \leq 0.0$		-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.96 \leq 0.0 \leq -1.0$			-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.92 \leq 0.0$		-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.92 \leq 0.0$		-0.8	-0.1	0.0	-0.3	7.8	> 0

Table A.2: **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table A.1.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma=60$, $c=4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma=60$, $c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma=30$, $c=4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma=30$, $c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma=60$, $c=4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma=60$, $c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma=30$, $c=4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma=30$, $c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma=60$, $c=4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma=60$, $c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma=30$, $c=4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma=30$, $c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table A.3: **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table A.1; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60$, $c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60$, $c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60$, $c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60$, $c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60$, $c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60$, $c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60$, $c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60$, $c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30$, $c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30$, $c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30$, $c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60$, $c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60$, $c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60$, $c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60$, $c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60$, $c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30$, $c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30$, $c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30$, $c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17

	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma = 60$, $c = 1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma = 60$, $c = 4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma = 60$, $c = 4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma = 60$, $c = 36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma = 30$, $c = 1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma = 30$, $c = 4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma = 30$, $c = 36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01